Plot Compendium

Photon Shower Shape Cell-Energy Reweighting Studies
MC23e $Z \to ee\gamma$ (DSID 700770)

ATLAS Simulation

March 13, 2026

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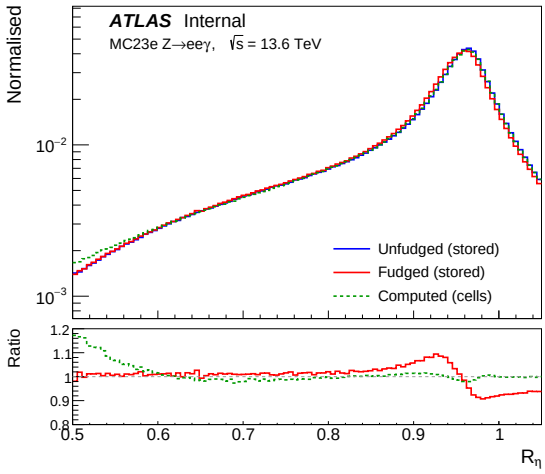
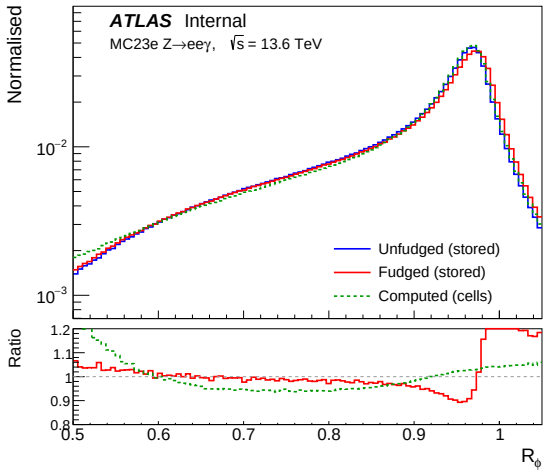
1 Initial Validation Plots

These plots validate the cell-level recomputation of the two shower shape variables (R_η, R_ϕ) against the stored values in the ntuples. Each figure shows three normalised distributions: **Unfudged/stored** (blue) — the original MC stored values; **Fudged/stored** (red) — the "fudged" stored values; and **Computed from cells** (green, dashed) — values recomputed from the 7×11 Layer-2 cell energies. The lower panel displays the ratio to the Unfudged baseline. Agreement between the blue and green curves confirms correct cell-level recomputation.

1.1 $Z \to ee\gamma$ channel



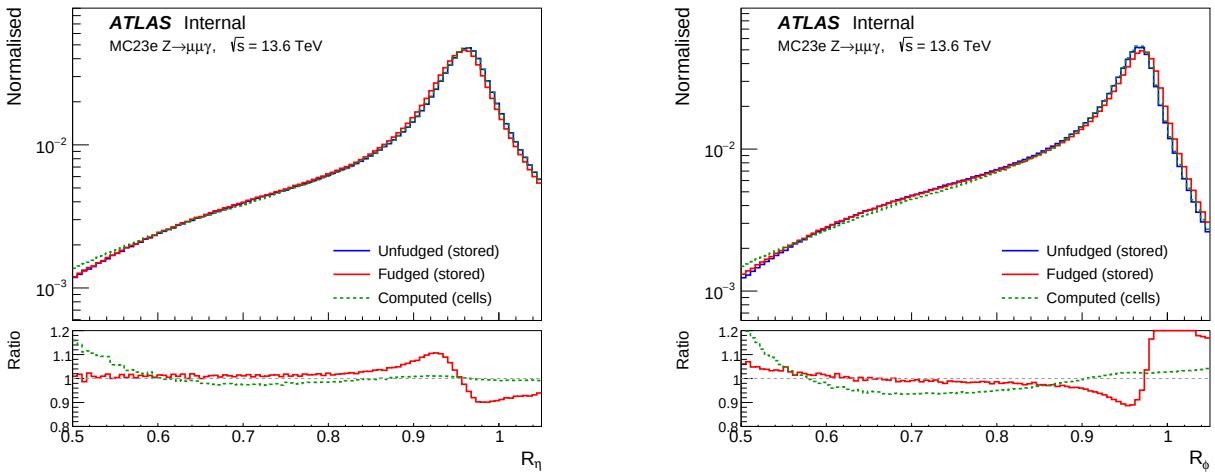
(a) R_η



(b) R_ϕ

Figure 1: Shower shape validation for the $Z \to ee\gamma$ channel. Normalised distributions of R_η and R_ϕ comparing stored (Unfudged, Fudged) and cell-recomputed values. The ratio panel confirms that the cell-level recomputation reproduces the stored values.

1.2 $Z \to \mu\mu\gamma$ channel

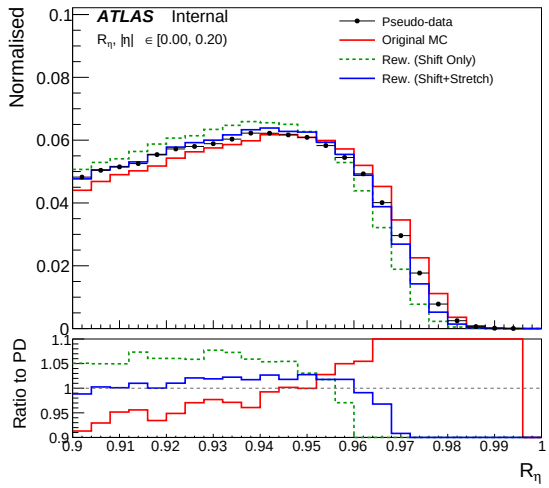


(a) R_η (b) R_ϕ

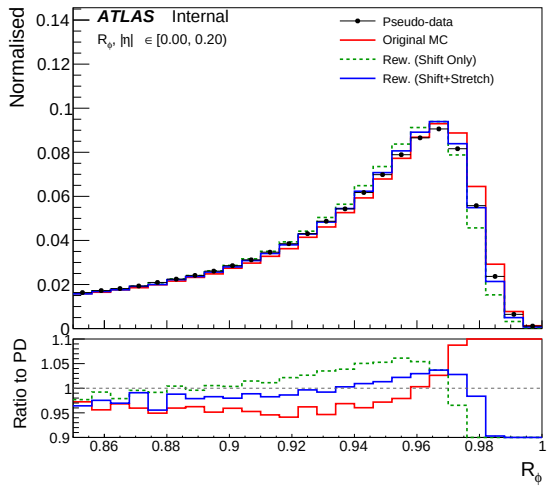
Figure 2: Shower shape validation for the $Z \to \mu\mu\gamma$ channel. Same format as fig. 1. The $Z \to \mu\mu\gamma$ sample provides an independent cross-check of the cell-level recomputation.

2 Closure Test — 5% Distortion ($\delta = 0.05$)

Closure test results with a 5% uniform distortion applied to all Layer-2 cell energies. Each figure shows one $|\eta|$ bin with both shower shape variables side by side. Upper panels: normalised distributions of pseudo-data, uncorrected MC, Method 1 (flat shift), and Method 2 (TProfile). Lower panels: ratio to pseudo-data.

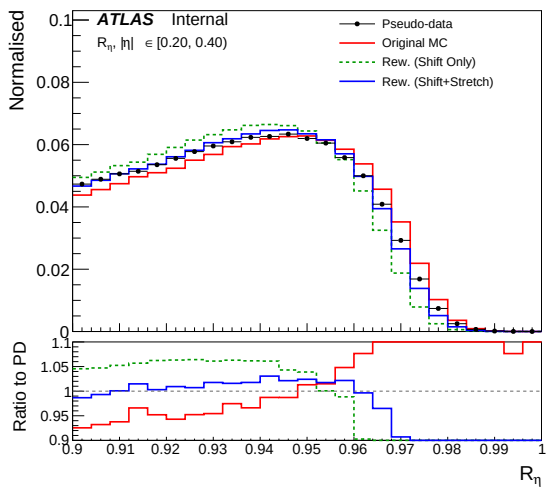


(a) R_η

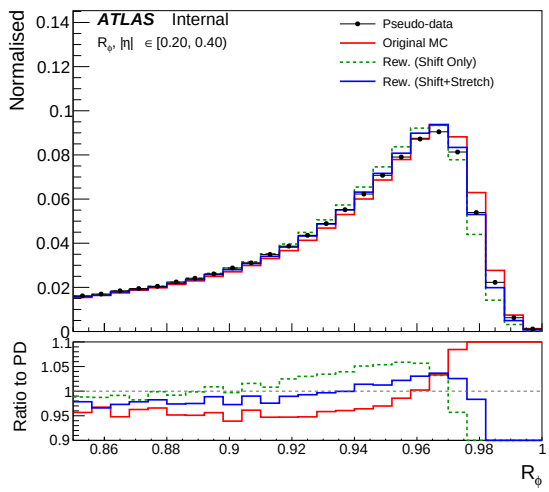


(b) R_ϕ

Figure 3: $|\eta| \in [0.00, 0.20)$, $\delta = 0.05$.

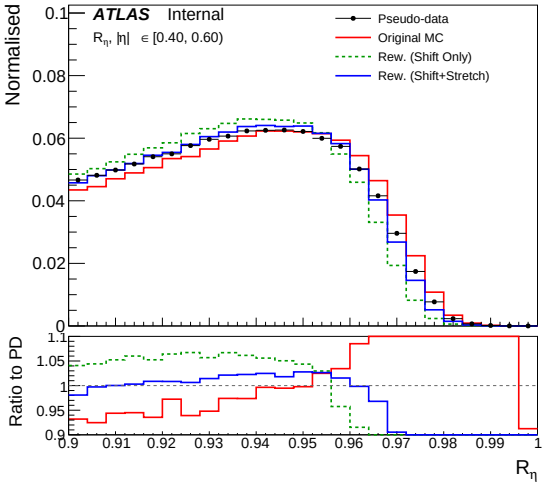


(a) R_η

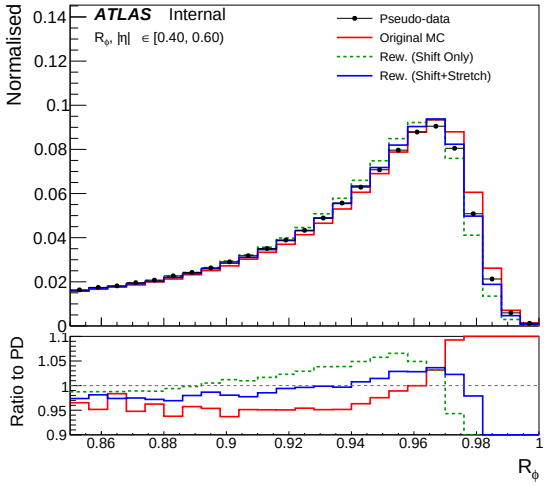


(b) R_ϕ

Figure 4: $|\eta| \in [0.20, 0.40)$, $\delta = 0.05$.

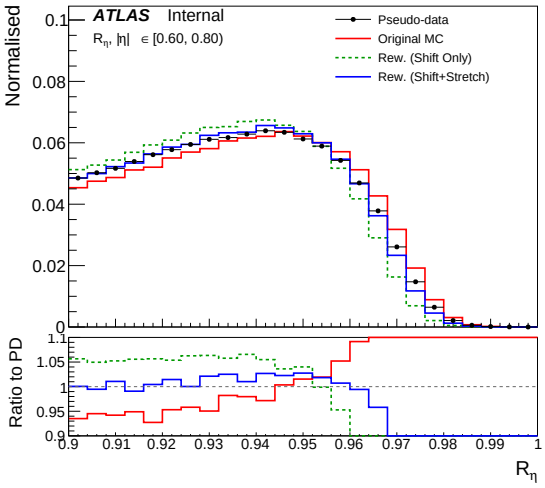


(a) R_η

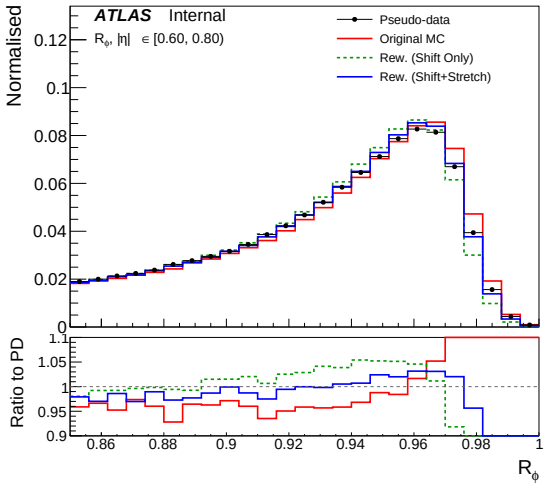


(b) R_ϕ

Figure 5: $|\eta| \in [0.40, 0.60)$, $\delta = 0.05$.

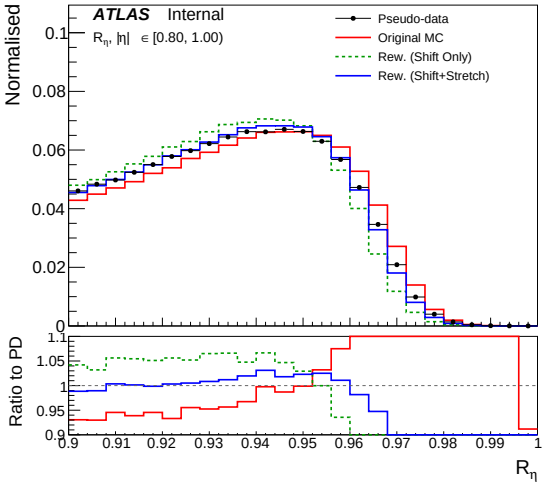


(a) R_η

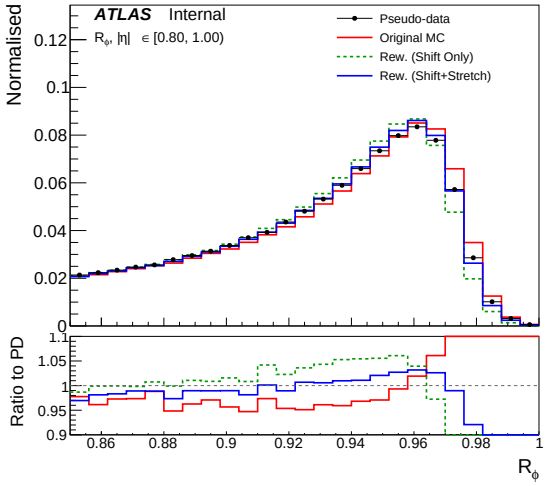


(b) R_ϕ

Figure 6: $|\eta| \in [0.60, 0.80)$, $\delta = 0.05$.

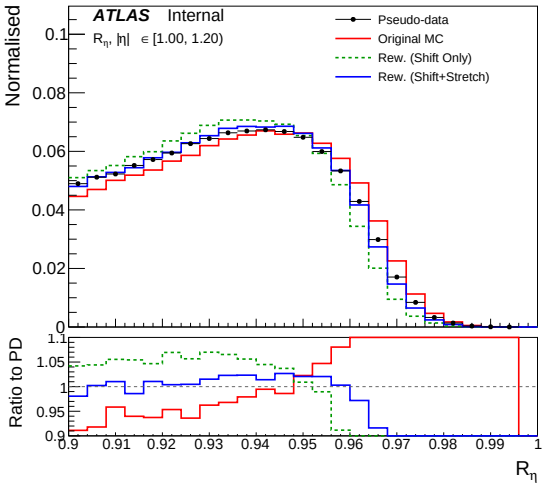


(a) R_η

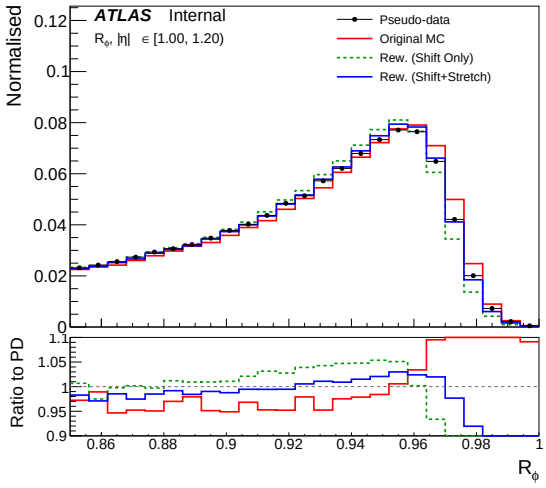


(b) R_ϕ

Figure 7: $|\eta| \in [0.80,\ 1.00)$, $\delta = 0.05$.

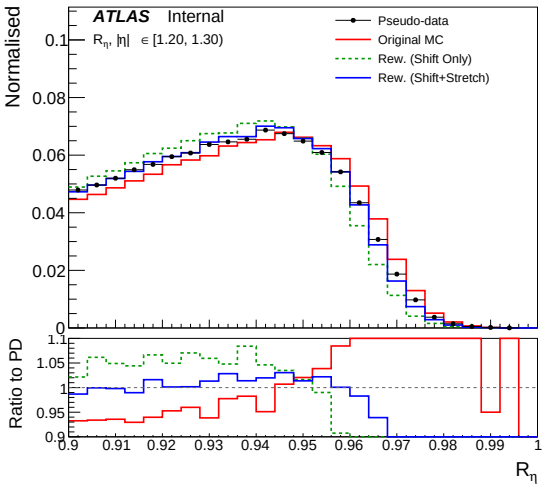


(a) R_η

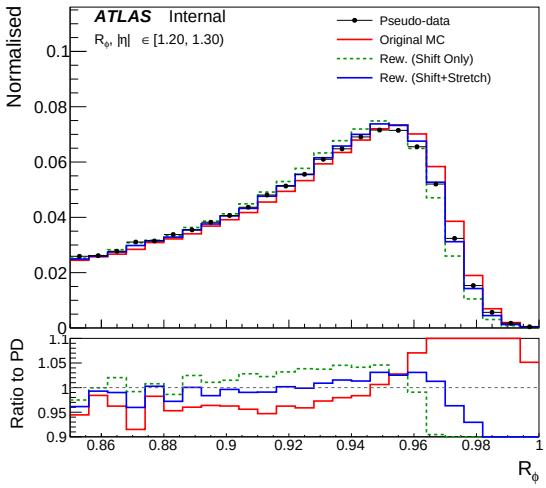


(b) R_ϕ

Figure 8: $|\eta| \in [1.00,\ 1.20)$, $\delta = 0.05$.

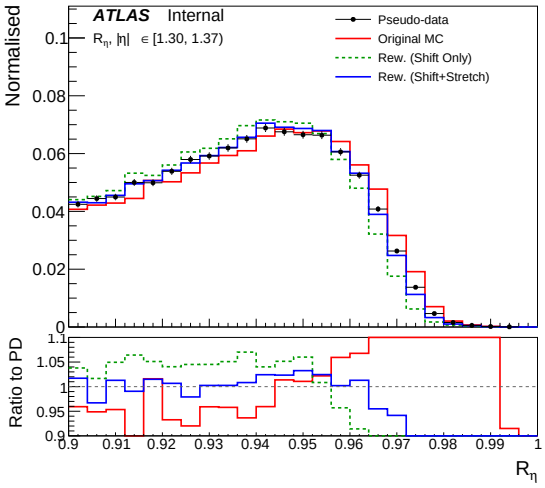


(a) R_η

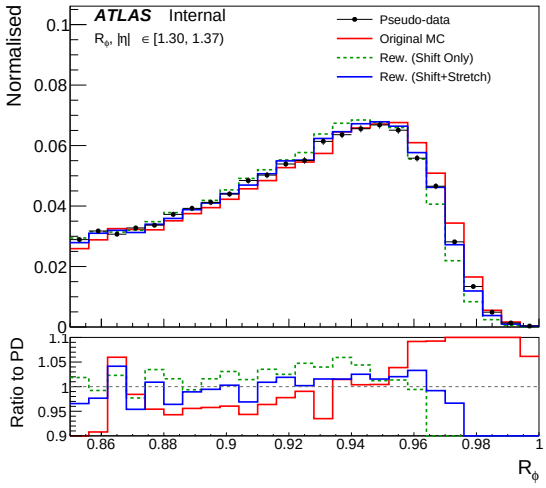


(b) R_ϕ

Figure 9: $|\eta| \in [1.20, 1.30)$, $\delta = 0.05$.

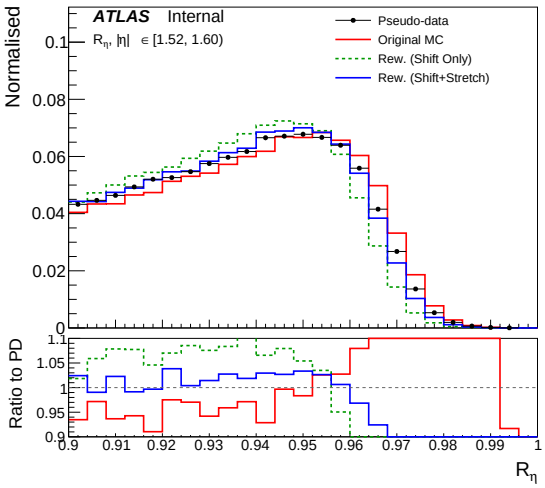


(a) R_η

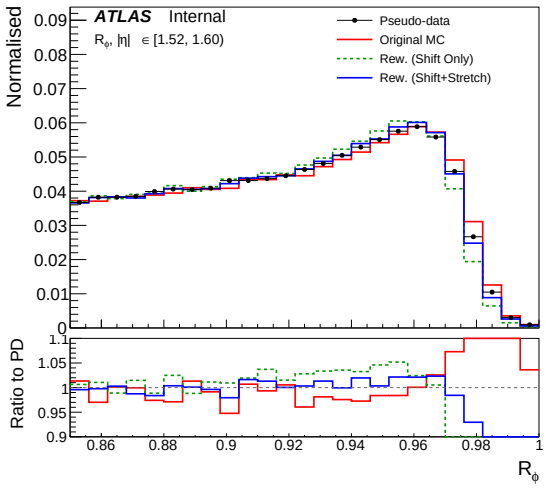


(b) R_ϕ

Figure 10: $|\eta| \in [1.30, 1.37)$, $\delta = 0.05$.

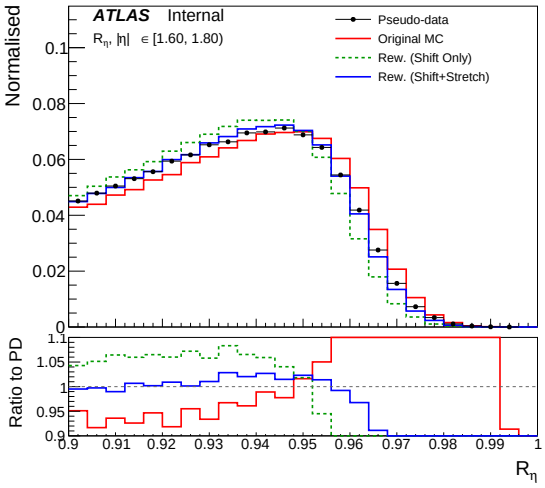


(a) R_η

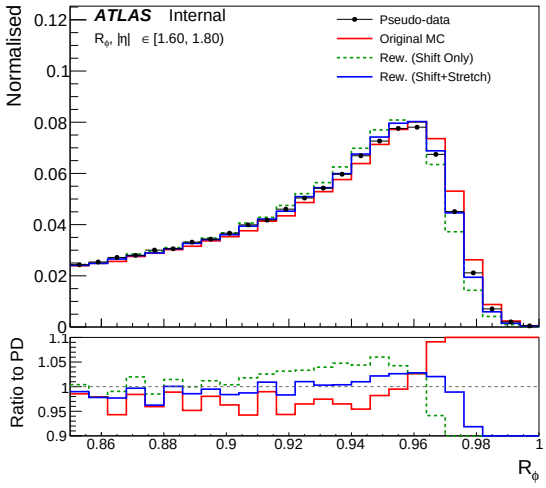


(b) R_ϕ

Figure 11: $|\eta| \in [1.52, 1.60)$, $\delta = 0.05$.

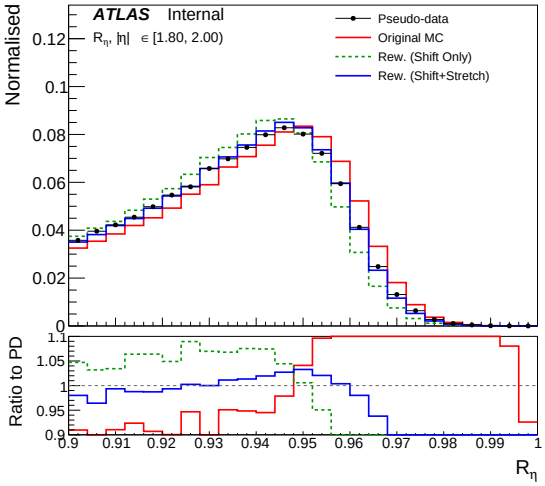


(a) R_η

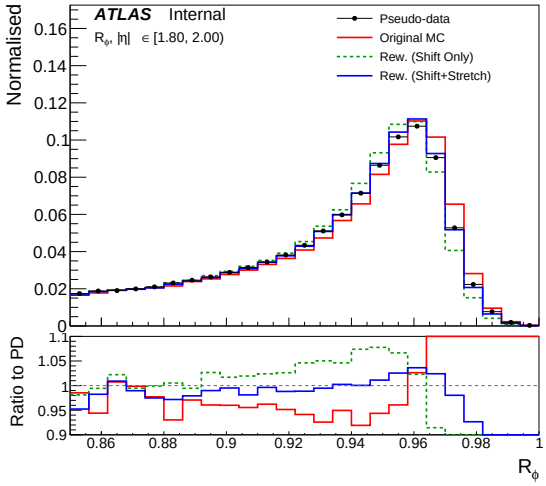


(b) R_ϕ

Figure 12: $|\eta| \in [1.60, 1.80)$, $\delta = 0.05$.

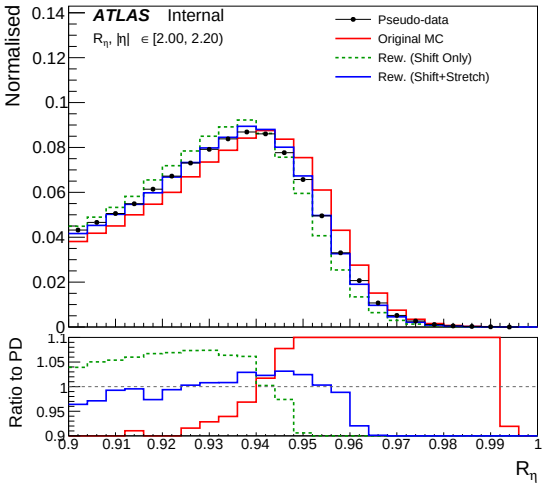


(a) R_η

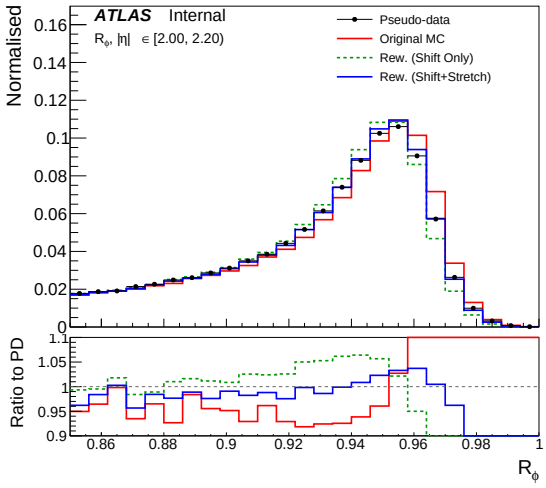


(b) R_ϕ

Figure 13: $|\eta| \in [1.80, 2.00)$, $\delta = 0.05$.

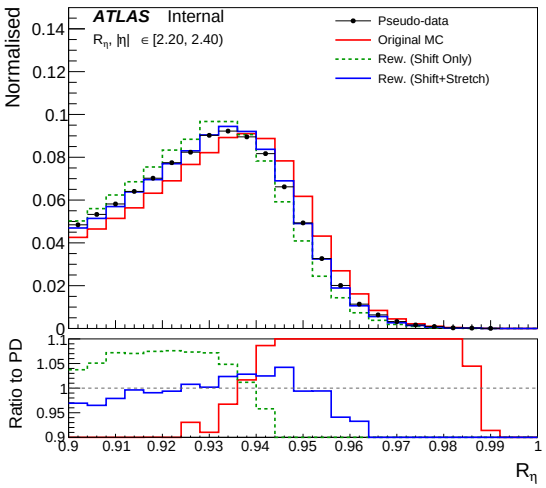


(a) R_η

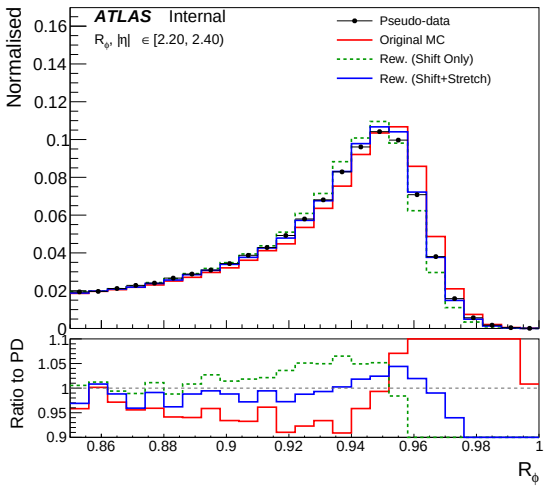


(b) R_ϕ

Figure 14: $|\eta| \in [2.00, 2.20)$, $\delta = 0.05$.



(a) R_η

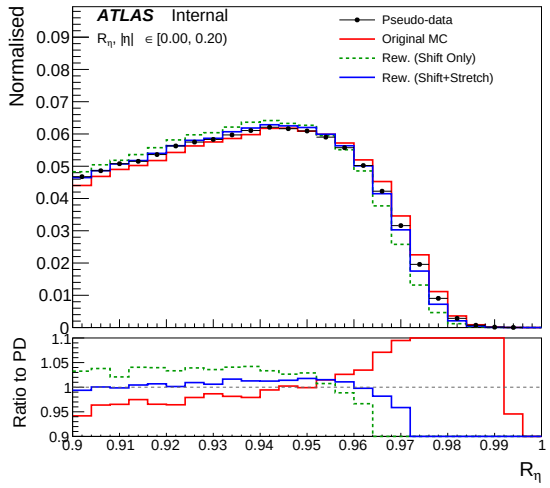


(b) R_ϕ

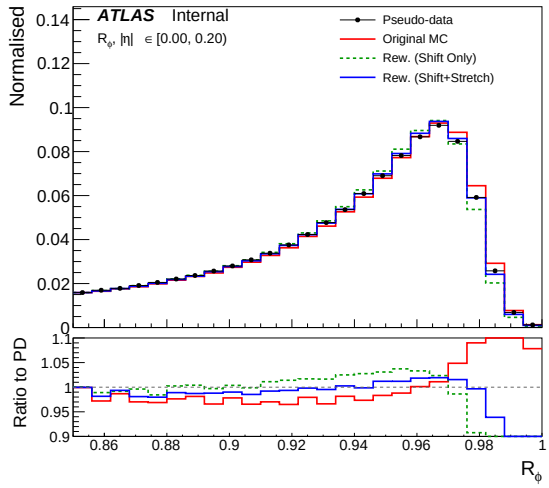
Figure 15: $|\eta| \in [2.20, 2.40)$, $\delta = 0.05$.

3 Closure Test — 3% Distortion ($\delta = 0.03$)

Closure test results with a 3% uniform distortion. Same format as section 2.

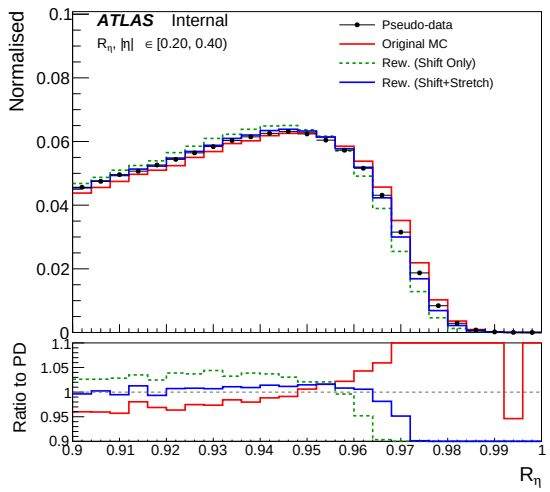


(a) R_η

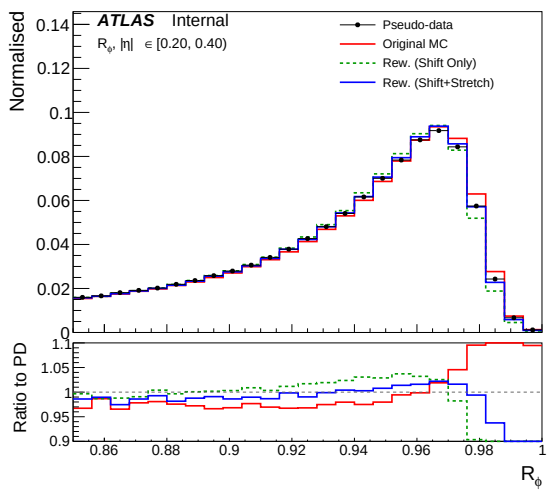


(b) R_ϕ

Figure 16: $|\eta| \in [0.00, 0.20)$, $\delta = 0.03$.

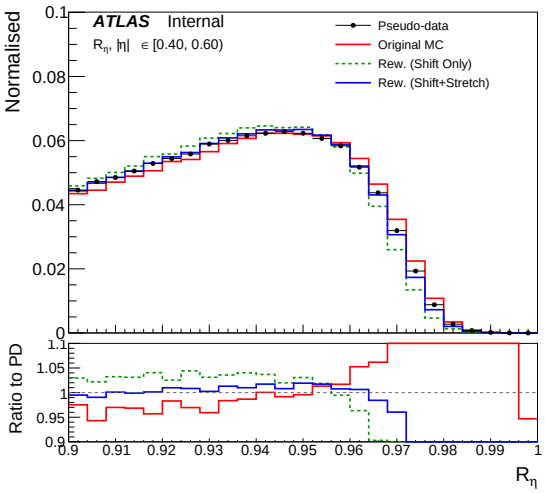


(a) R_η

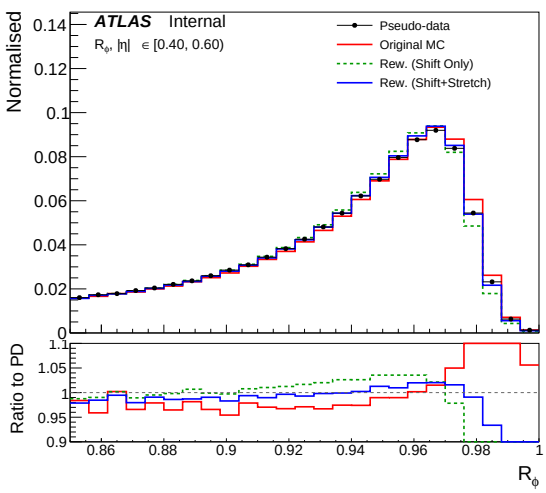


(b) R_ϕ

Figure 17: $|\eta| \in [0.20, 0.40)$, $\delta = 0.03$.

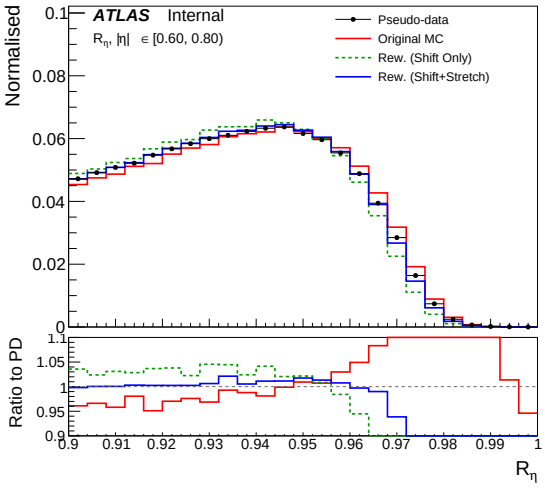


(a) R_η

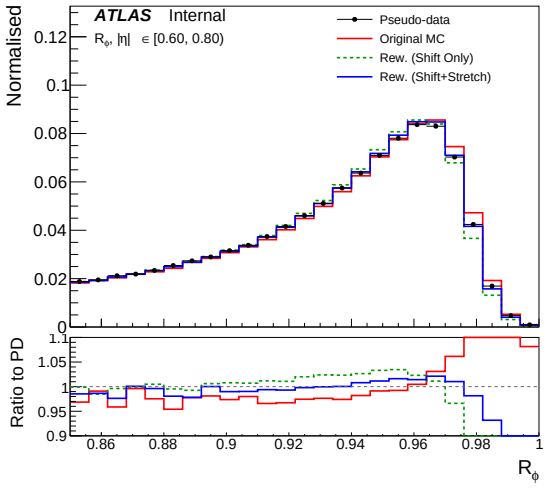


(b) R_ϕ

Figure 18: $|\eta| \in [0.40,\ 0.60)$, $\delta = 0.03$.

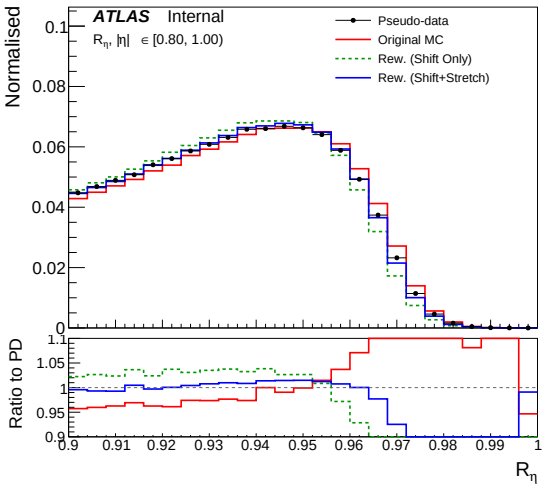


(a) R_η

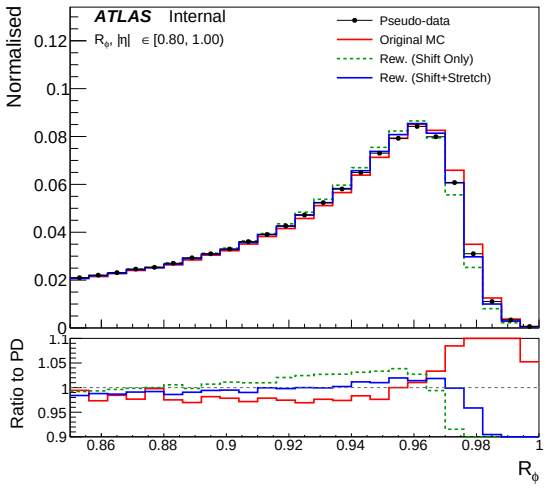


(b) R_ϕ

Figure 19: $|\eta| \in [0.60,\ 0.80)$, $\delta = 0.03$.

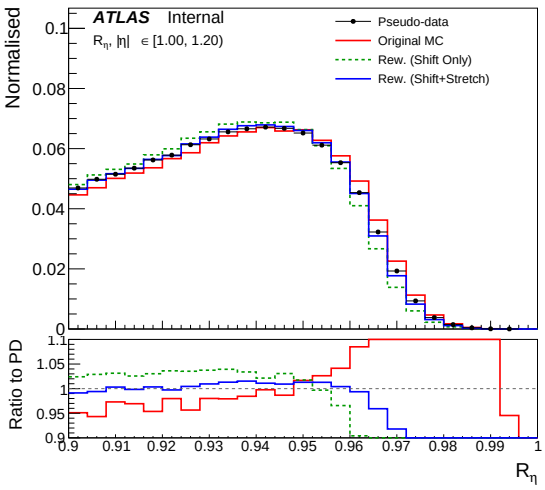


(a) R_η

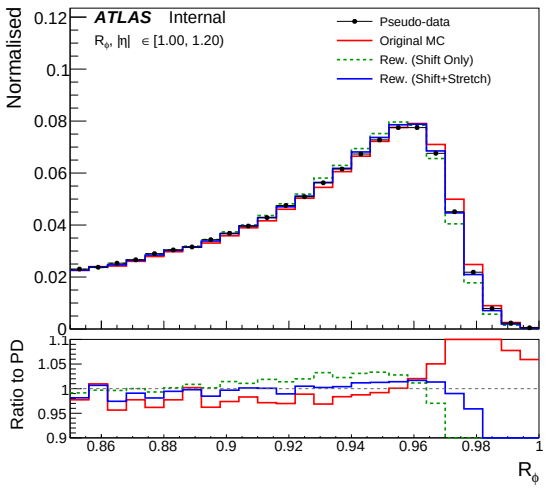


(b) R_ϕ

Figure 20: $|\eta| \in [0.80,\ 1.00)$, $\delta = 0.03$.

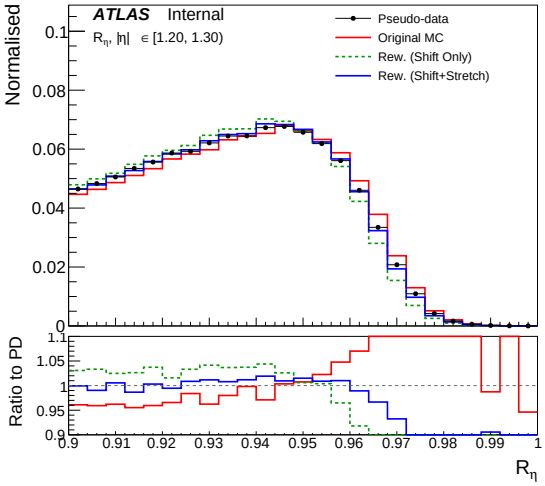


(a) R_η

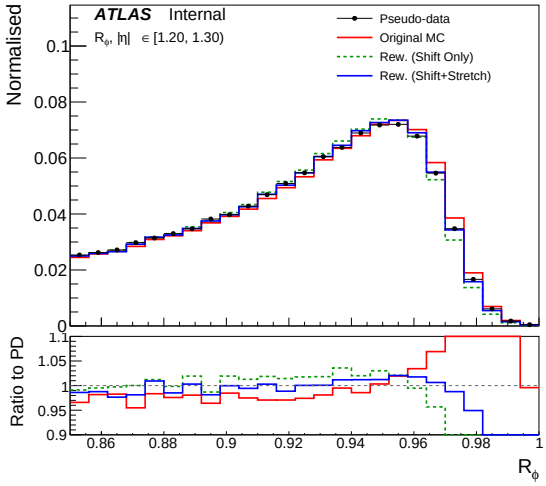


(b) R_ϕ

Figure 21: $|\eta| \in [1.00,\ 1.20)$, $\delta = 0.03$.

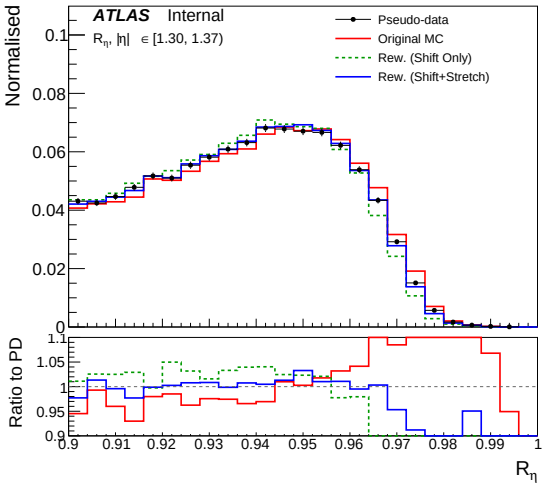


(a) R_η

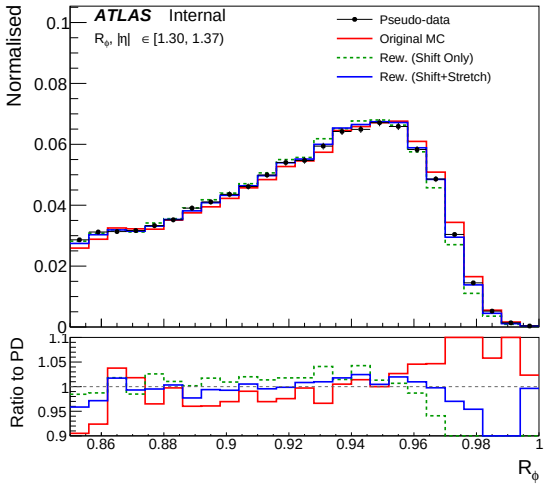


(b) R_ϕ

Figure 22: $|\eta| \in [1.20, 1.30)$, $\delta = 0.03$.

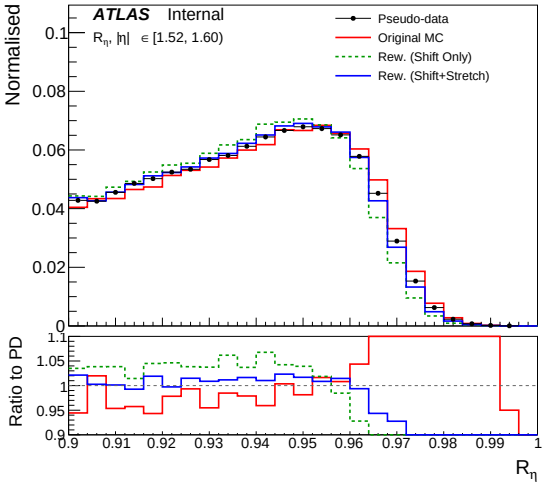


(a) R_η

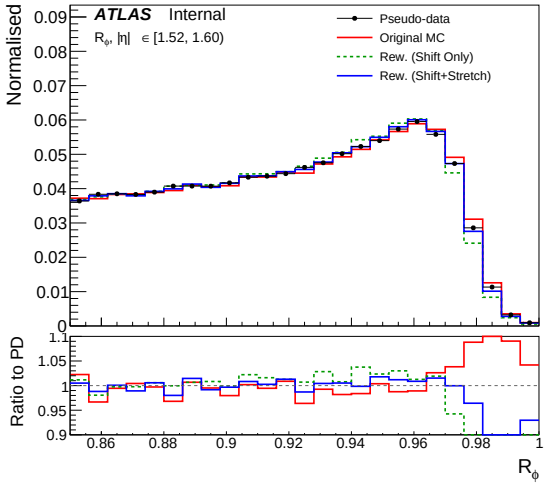


(b) R_ϕ

Figure 23: $|\eta| \in [1.30, 1.37)$, $\delta = 0.03$.

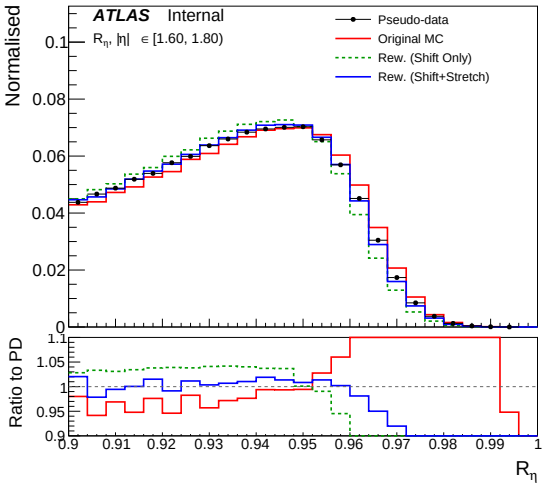


(a) R_η

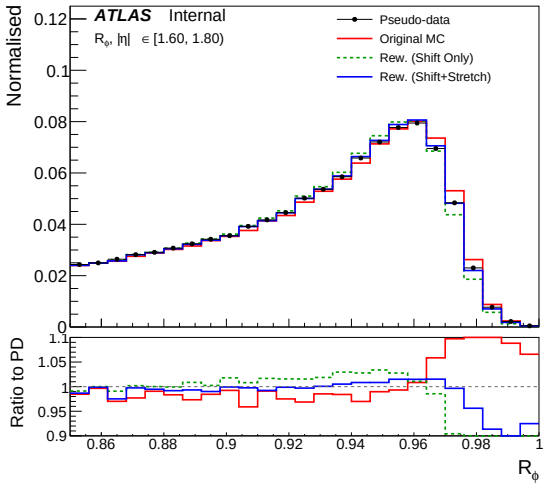


(b) R_ϕ

Figure 24: $|\eta| \in [1.52,\ 1.60)$, $\delta = 0.03$.

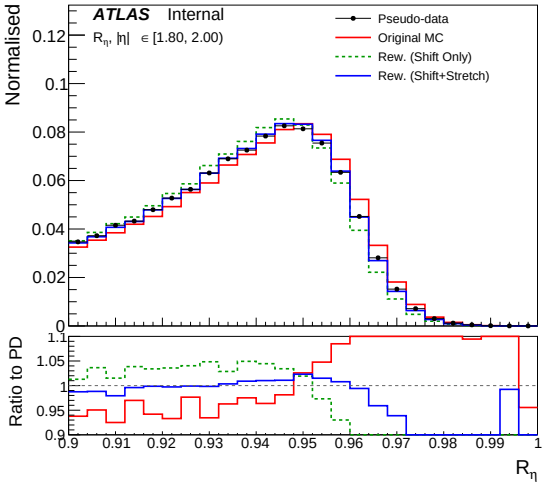


(a) R_η

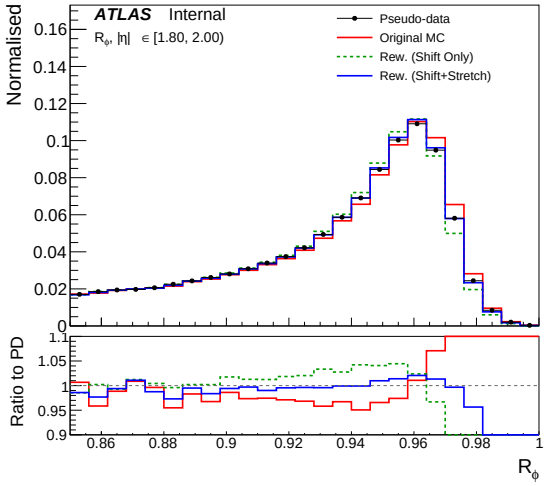


(b) R_ϕ

Figure 25: $|\eta| \in [1.60,\ 1.80)$, $\delta = 0.03$.

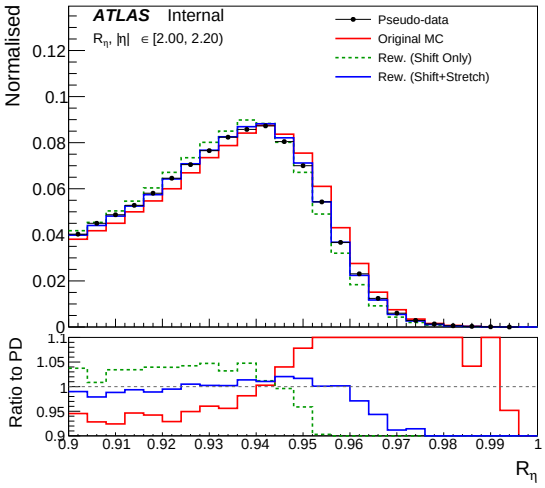


(a) R_η

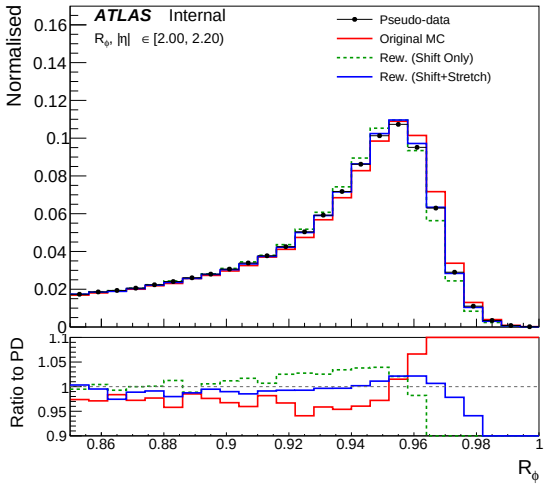


(b) R_ϕ

Figure 26: $|\eta| \in [1.80, 2.00)$, $\delta = 0.03$.

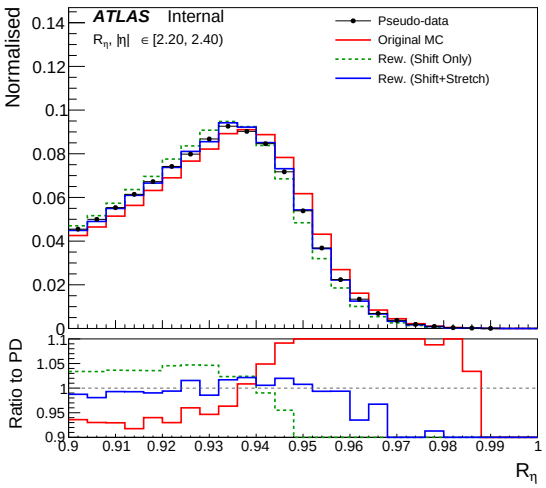


(a) R_η

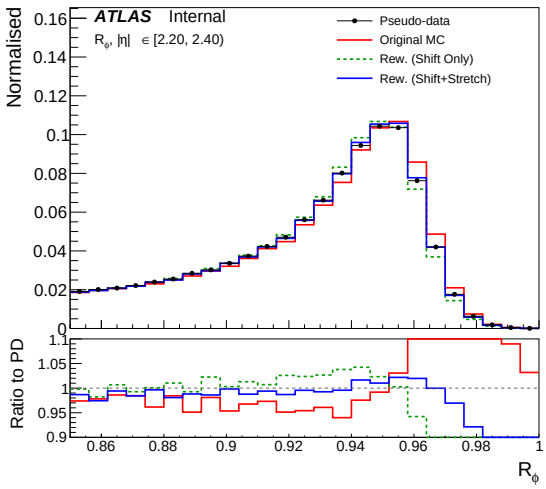


(b) R_ϕ

Figure 27: $|\eta| \in [2.00, 2.20)$, $\delta = 0.03$.



(a) R_η

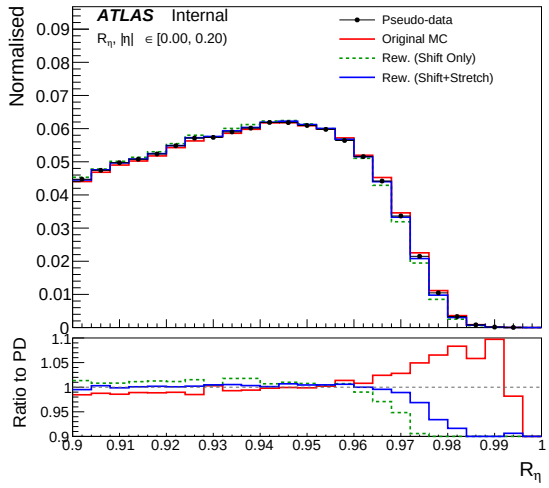


(b) R_ϕ

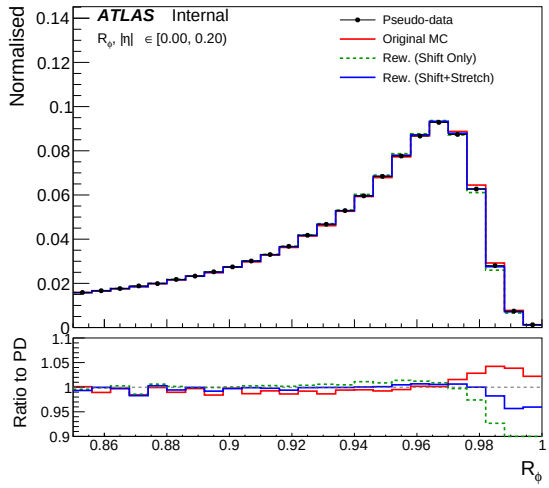
Figure 28: $|\eta| \in [2.20, 2.40)$, $\delta = 0.03$.

4 Closure Test — 1% Distortion ($\delta = 0.01$)

Closure test results with a 1% uniform distortion. Same format as section 2.

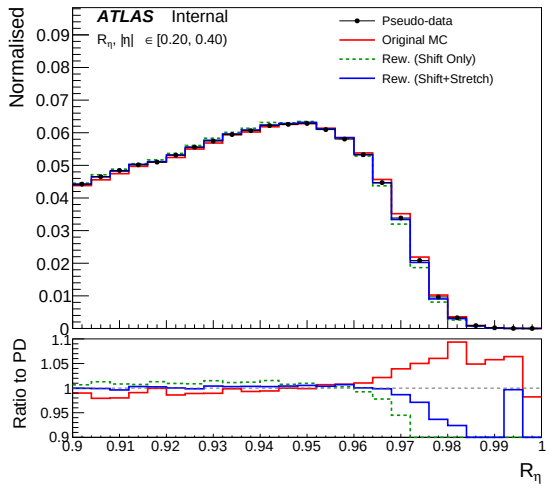


(a) R_η

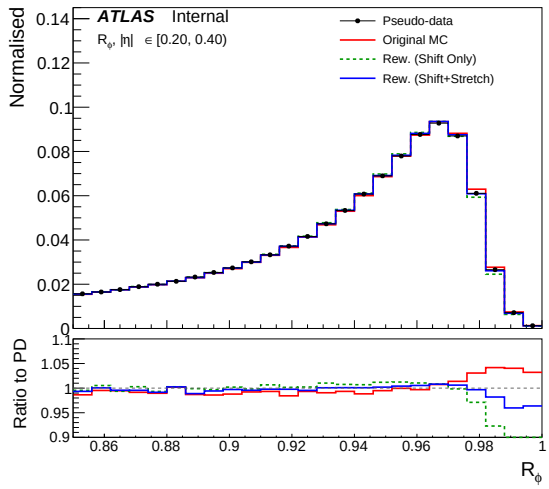


(b) R_ϕ

Figure 29: $|\eta| \in [0.00, 0.20)$, $\delta = 0.01$.

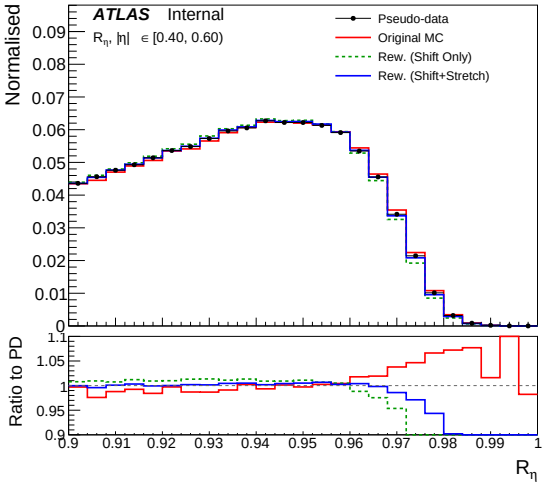


(a) R_η

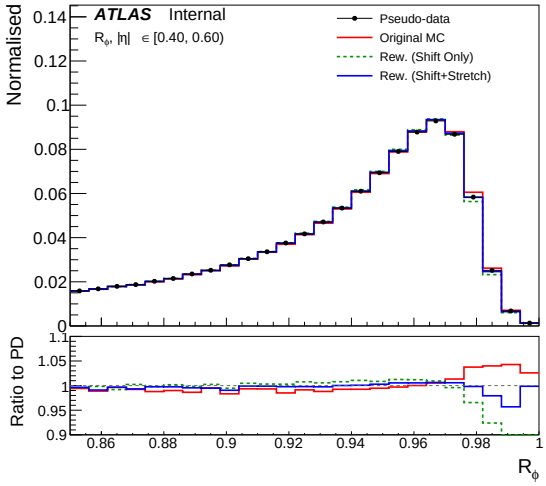


(b) R_ϕ

Figure 30: $|\eta| \in [0.20, 0.40)$, $\delta = 0.01$.

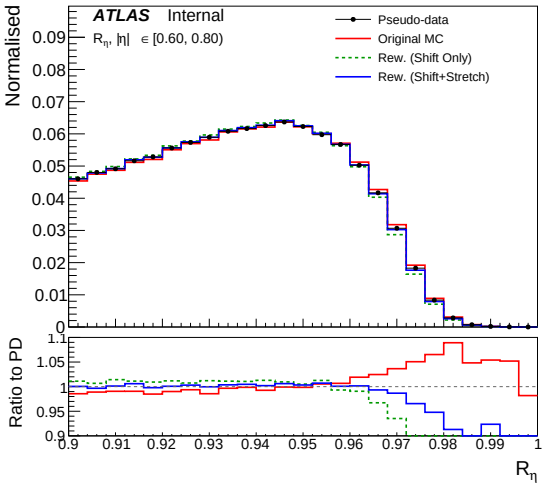


(a) R_η

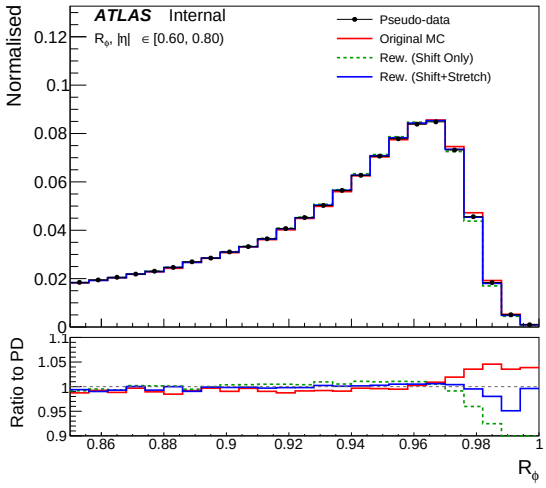


(b) R_ϕ

Figure 31: $|\eta| \in [0.40,\ 0.60)$, $\delta = 0.01$.

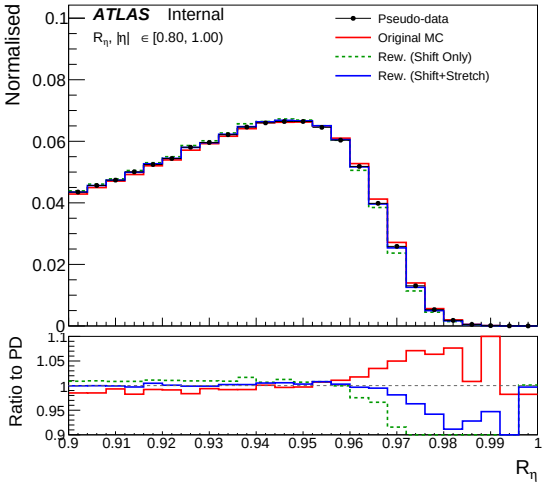


(a) R_η

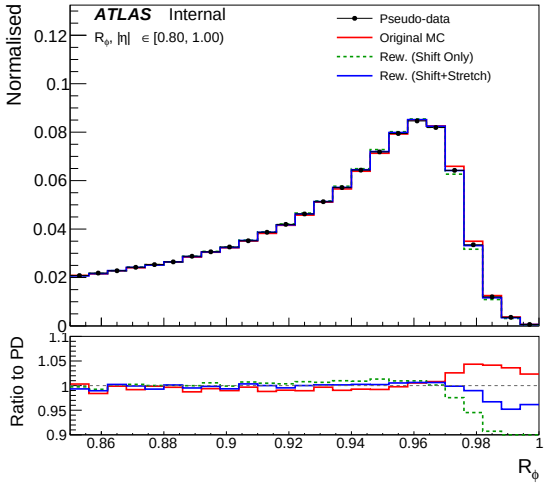


(b) R_ϕ

Figure 32: $|\eta| \in [0.60,\ 0.80)$, $\delta = 0.01$.

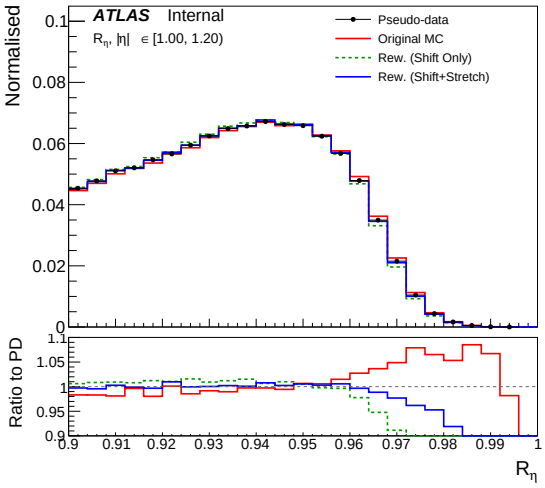


(a) R_η

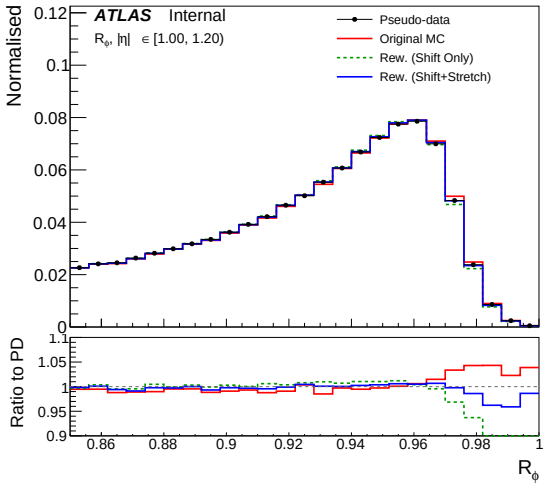


(b) R_ϕ

Figure 33: $|\eta| \in [0.80,\ 1.00)$, $\delta = 0.01$.

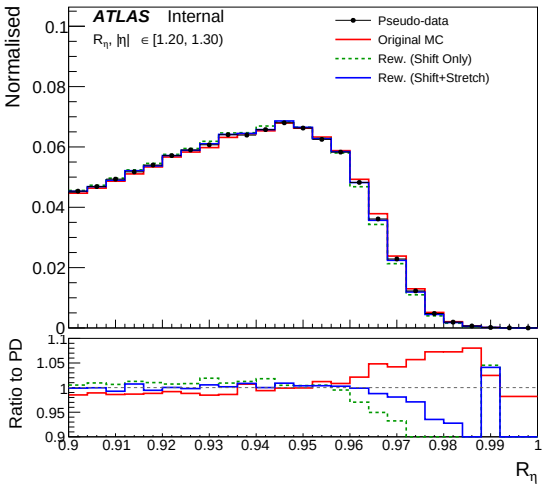


(a) R_η

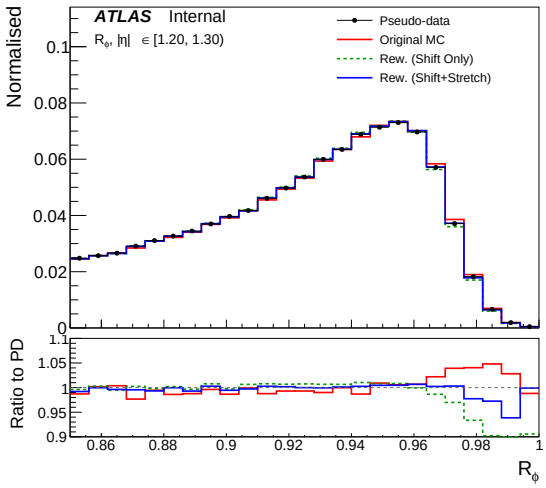


(b) R_ϕ

Figure 34: $|\eta| \in [1.00,\ 1.20)$, $\delta = 0.01$.

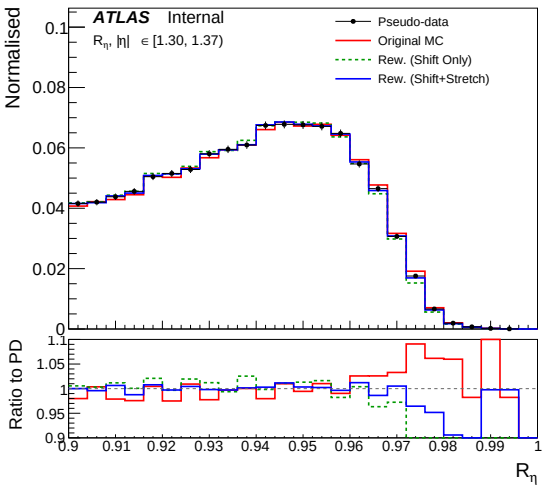


(a) R_η

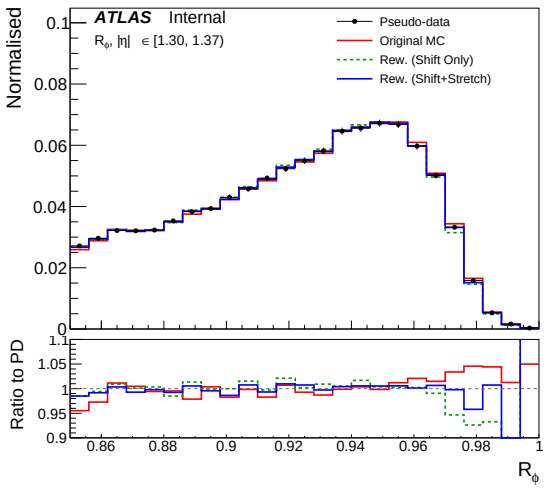


(b) R_ϕ

Figure 35: $|\eta| \in [1.20,\ 1.30)$, $\delta = 0.01$.

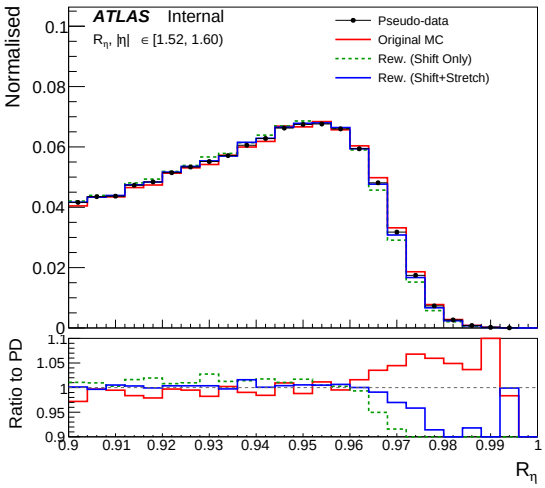


(a) R_η

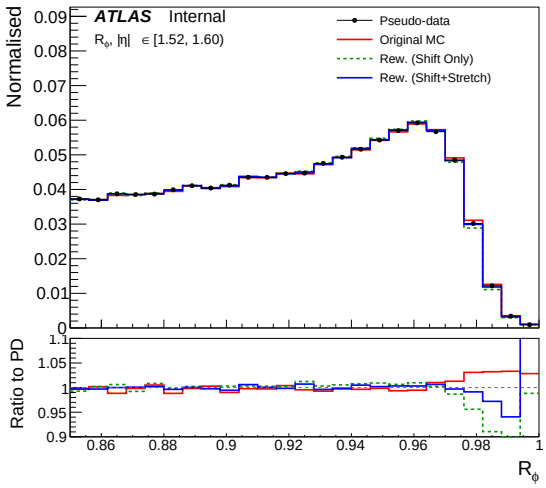


(b) R_ϕ

Figure 36: $|\eta| \in [1.30,\ 1.37)$, $\delta = 0.01$.

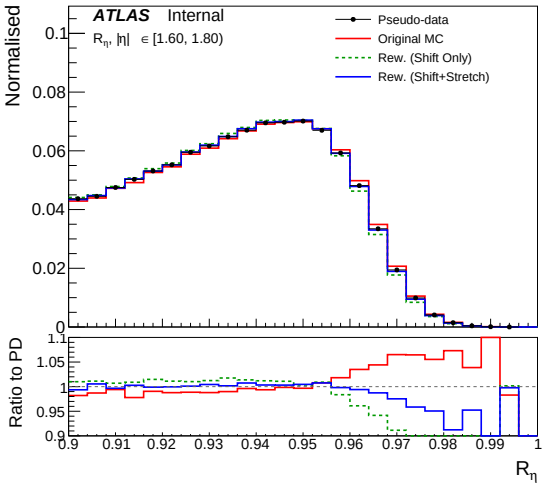


(a) R_η

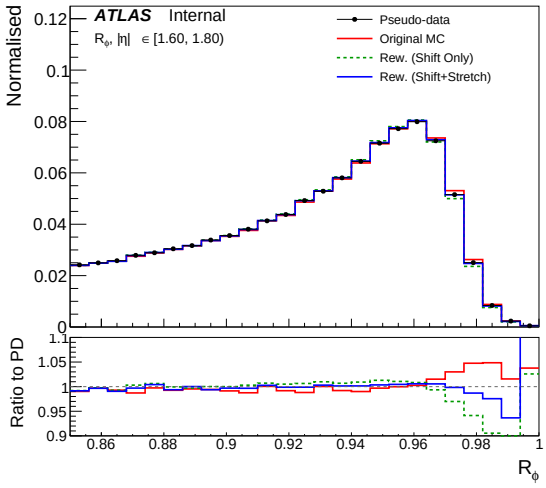


(b) R_ϕ

Figure 37: $|\eta| \in [1.52, 1.60)$, $\delta = 0.01$.

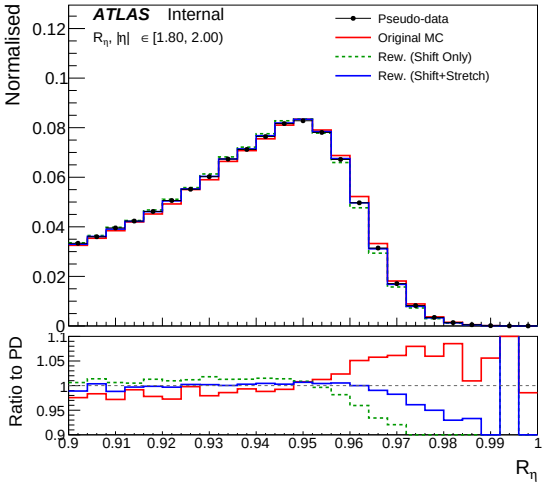


(a) R_η

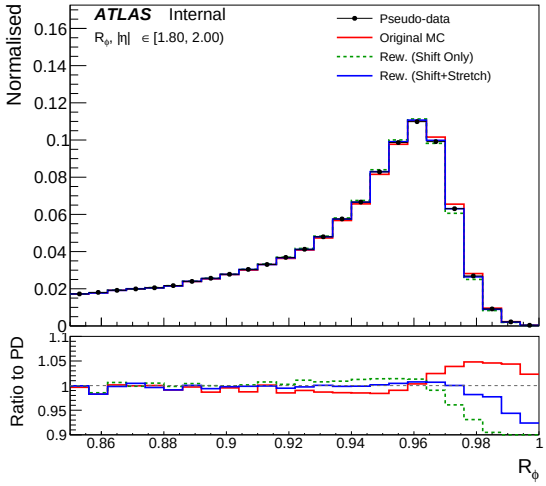


(b) R_ϕ

Figure 38: $|\eta| \in [1.60, 1.80)$, $\delta = 0.01$.

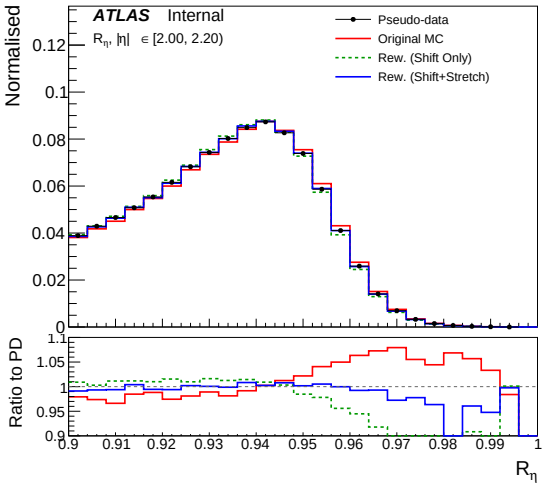


(a) R_η

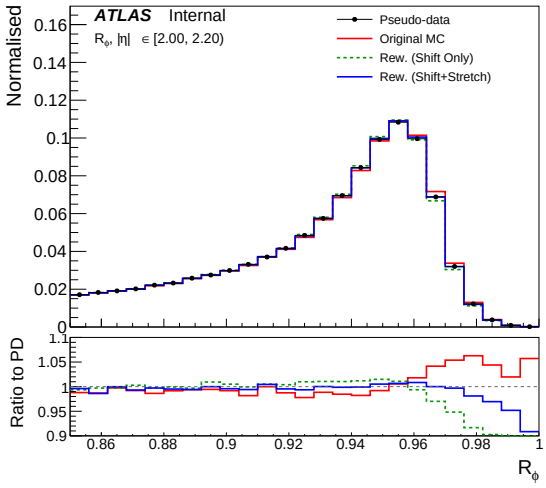


(b) R_ϕ

Figure 39: $|\eta| \in [1.80, 2.00)$, $\delta = 0.01$.

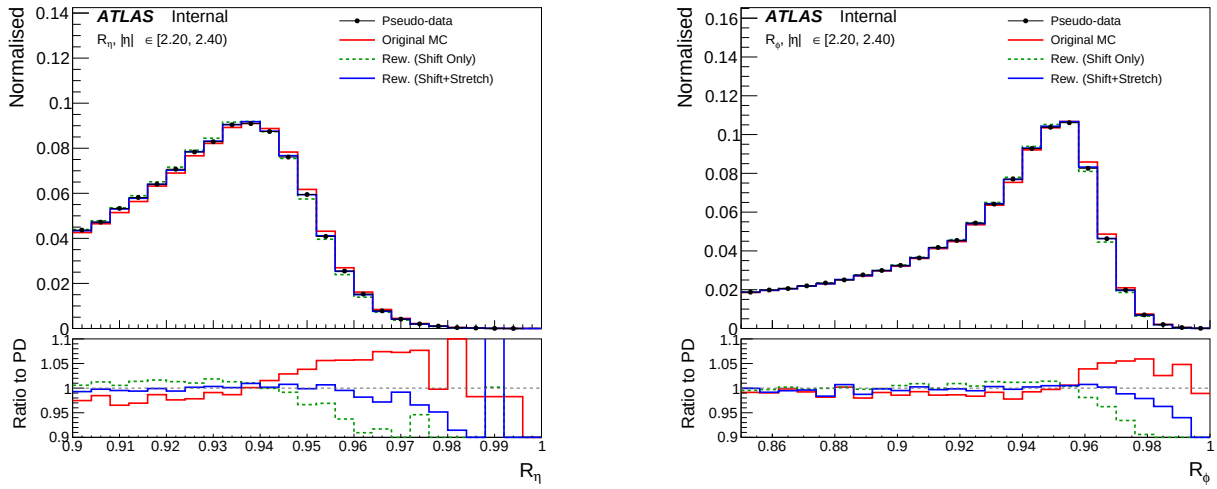


(a) R_η



(b) R_ϕ

Figure 40: $|\eta| \in [2.00, 2.20)$, $\delta = 0.01$.



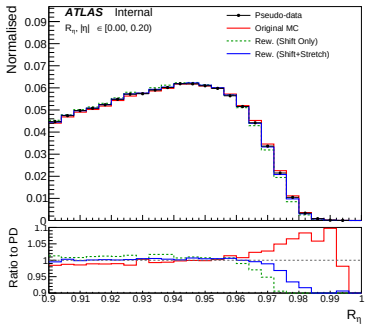
(a) R_η (b) R_ϕ

Figure 41: $|\eta| \in [2.20, 2.40)$, $\delta = 0.01$.

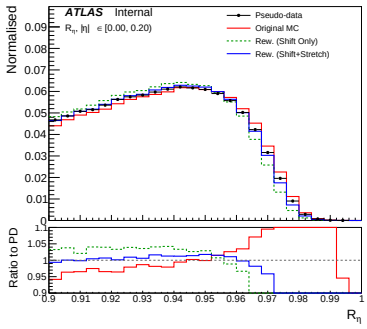
5 Distortion Level Comparison

Each figure shows the three distortion levels (1%, 3%, 5%) side by side for the same $|\eta|$ bin. This allows direct assessment of how correction performance scales with distortion magnitude.

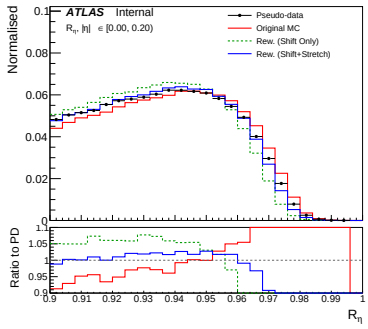
5.1 R_η



(a) $\delta = 1\%$

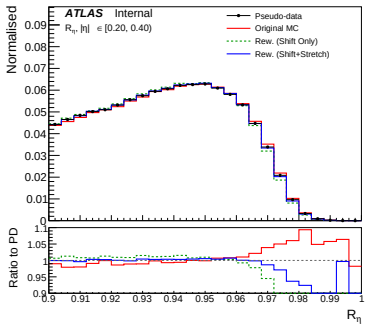


(b) $\delta = 3\%$

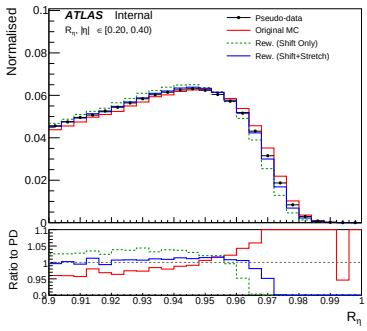


(c) $\delta = 5\%$

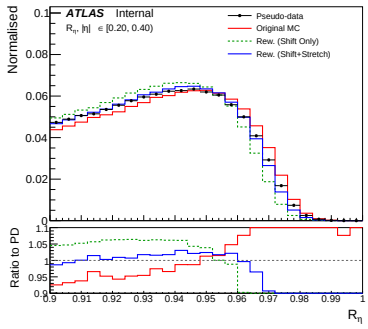
Figure 42: R_η, $|\eta| \in [0.00, 0.20)$.



(a) $\delta = 1\%$

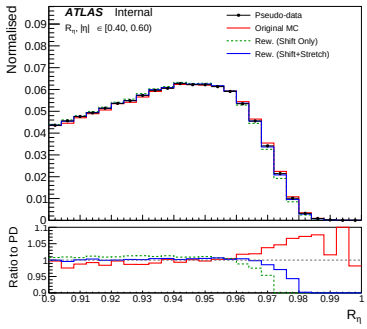


(b) $\delta = 3\%$

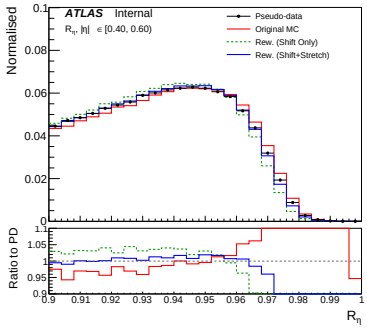


(c) $\delta = 5\%$

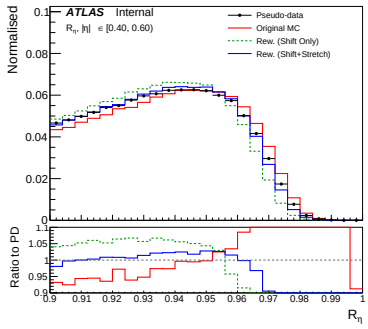
Figure 43: R_η, $|\eta| \in [0.20, 0.40)$.



(a) $\delta = 1\%$

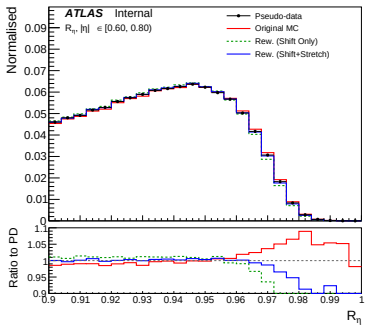


(b) $\delta = 3\%$

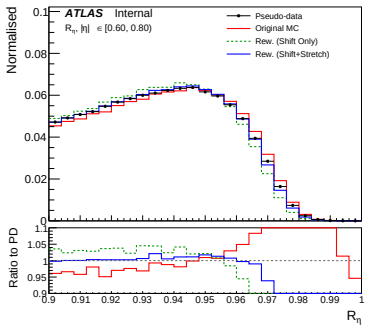


(c) $\delta = 5\%$

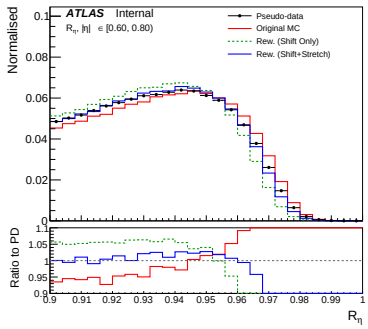
Figure 44: R_η, $|\eta| \in [0.40, 0.60)$.



(a) $\delta = 1\%$

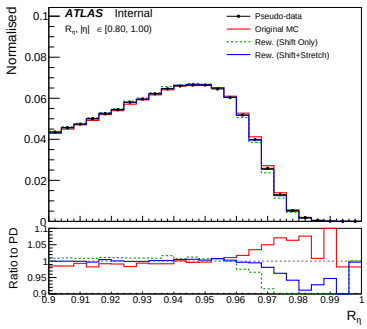


(b) $\delta = 3\%$

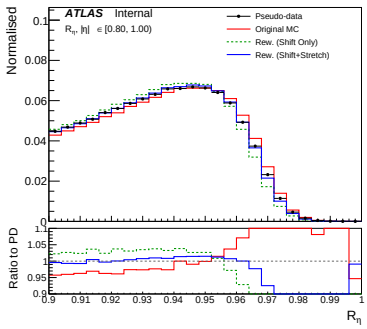


(c) $\delta = 5\%$

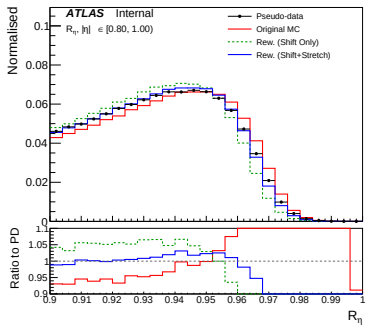
Figure 45: R_η, $|\eta| \in [0.60, 0.80)$.



(a) $\delta = 1\%$

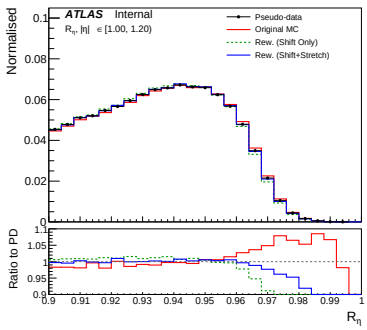


(b) $\delta = 3\%$

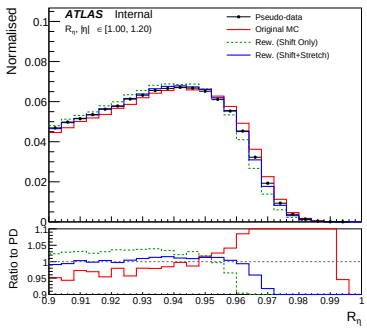


(c) $\delta = 5\%$

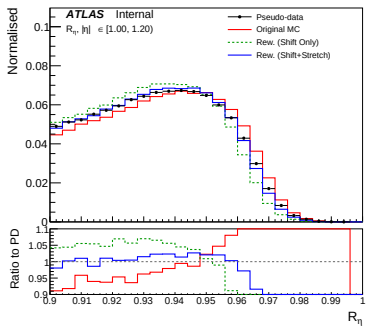
Figure 46: R_η, $|\eta| \in [0.80, 1.00)$.



(a) $\delta = 1\%$

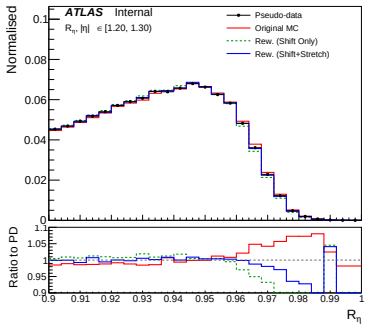


(b) $\delta = 3\%$

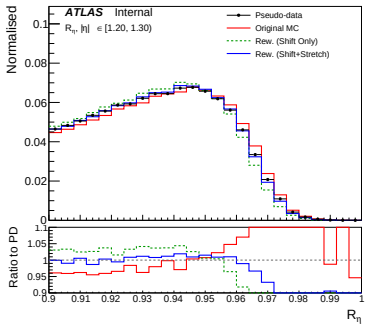


(c) $\delta = 5\%$

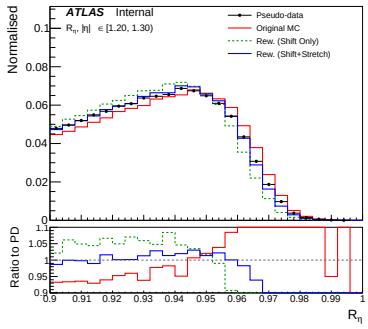
Figure 47: R_η, $|\eta| \in [1.00, 1.20)$.



(a) $\delta = 1\%$

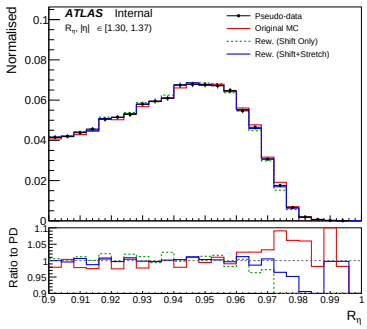


(b) $\delta = 3\%$

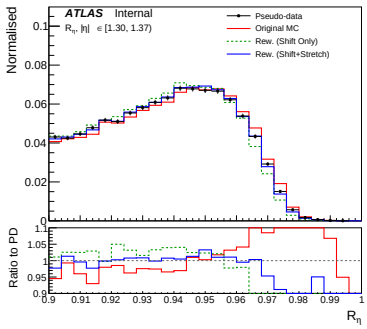


(c) $\delta = 5\%$

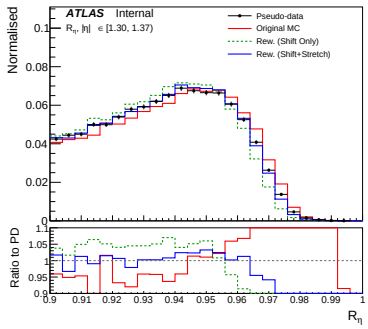
Figure 48: R_η, $|\eta| \in [1.20, 1.30)$.



(a) $\delta = 1\%$

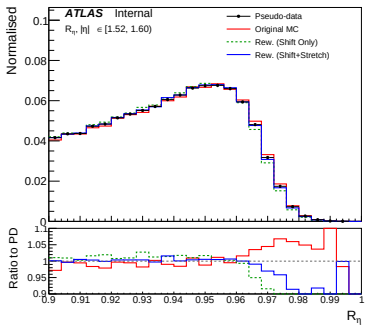


(b) $\delta = 3\%$

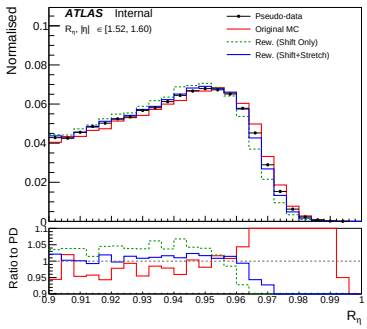


(c) $\delta = 5\%$

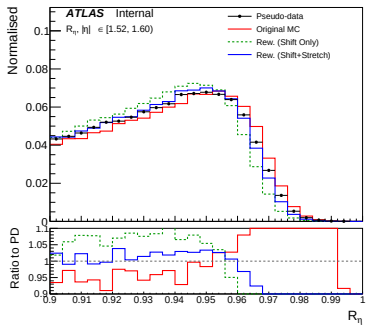
Figure 49: R_η, $|\eta| \in [1.30, 1.37)$.



(a) $\delta = 1\%$

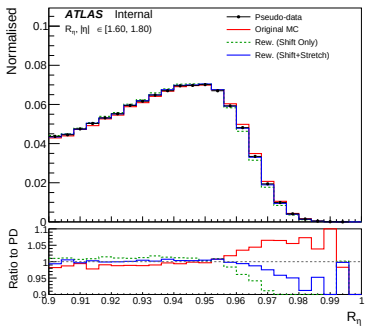


(b) $\delta = 3\%$

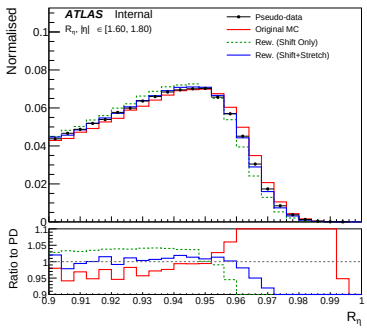


(c) $\delta = 5\%$

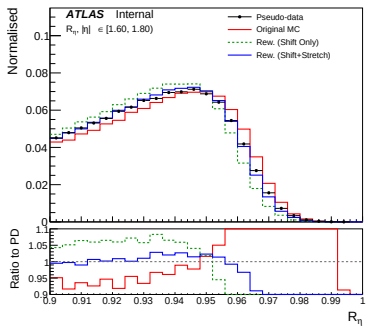
Figure 50: R_η, $|\eta| \in [1.52, 1.60)$.



(a) $\delta = 1\%$

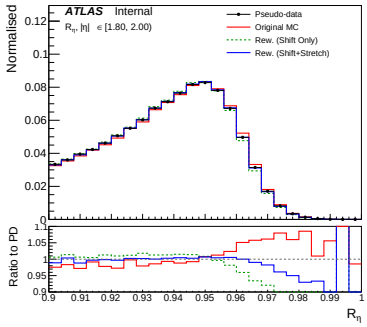


(b) $\delta = 3\%$

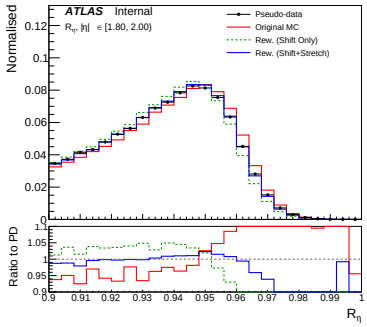


(c) $\delta = 5\%$

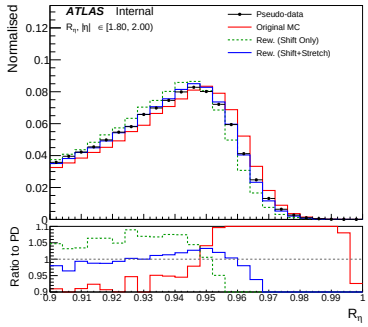
Figure 51: R_η, $|\eta| \in [1.60, 1.80)$.



(a) $\delta = 1\%$

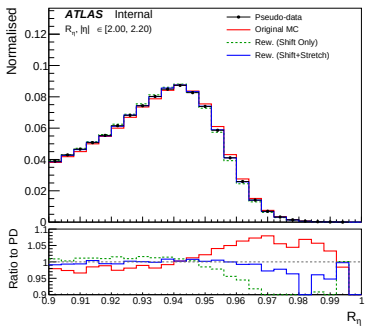


(b) $\delta = 3\%$

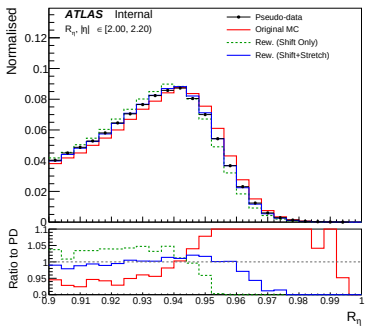


(c) $\delta = 5\%$

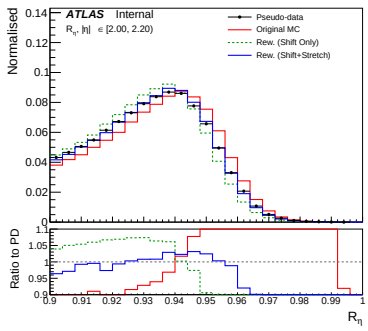
Figure 52: R_η, $|\eta| \in [1.80, 2.00)$.



(a) $\delta = 1\%$

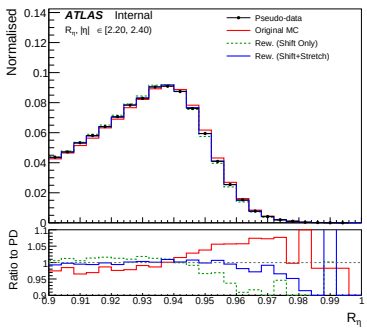


(b) $\delta = 3\%$

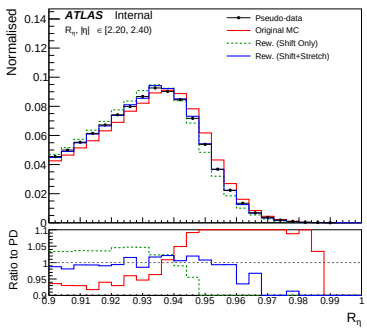


(c) $\delta = 5\%$

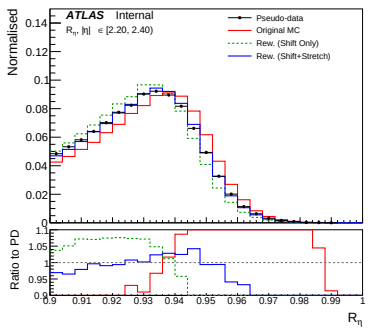
Figure 53: R_η, $|\eta| \in [2.00, 2.20)$.



(a) $\delta = 1\%$



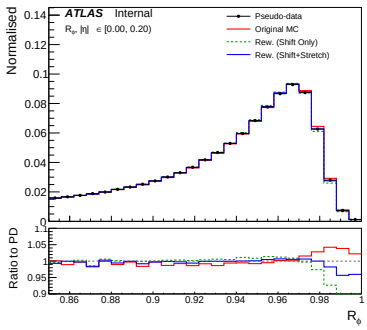
(b) $\delta = 3\%$



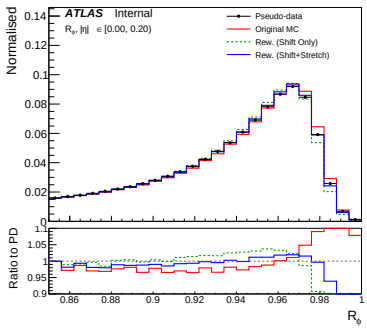
(c) $\delta = 5\%$

Figure 54: R_η, $|\eta| \in [2.20, 2.40)$.

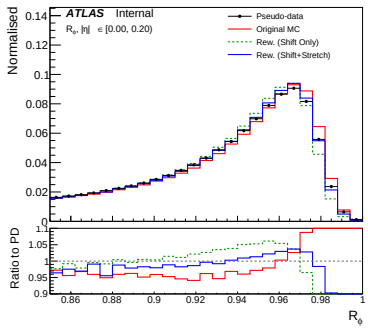
5.2 R_ϕ



(a) $\delta = 1\%$

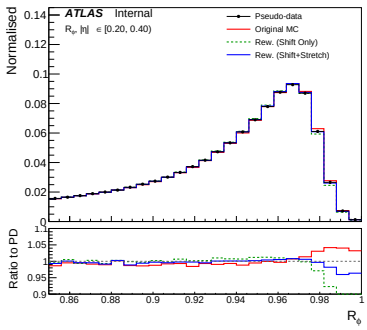


(b) $\delta = 3\%$

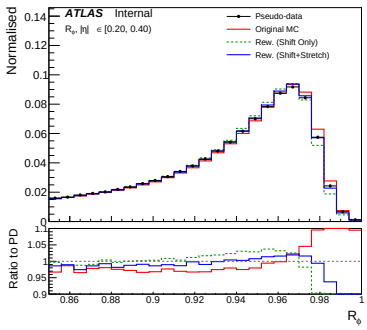


(c) $\delta = 5\%$

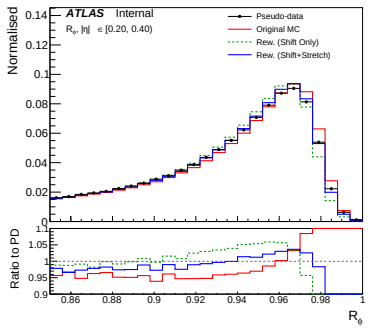
Figure 55: R_ϕ, $|\eta| \in [0.00, 0.20)$.



(a) $\delta = 1\%$

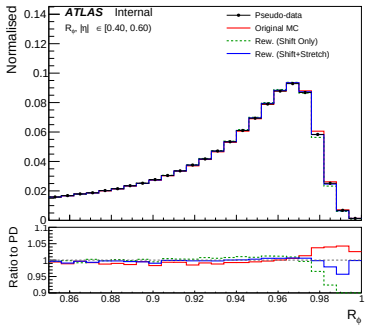


(b) $\delta = 3\%$

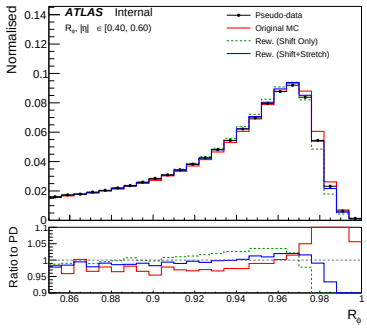


(c) $\delta = 5\%$

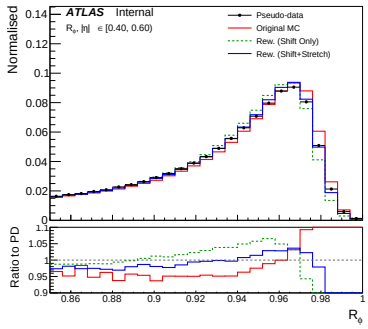
Figure 56: R_ϕ, $|\eta| \in [0.20, 0.40)$.



(a) $\delta = 1\%$

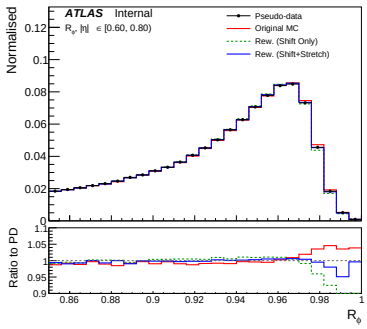


(b) $\delta = 3\%$

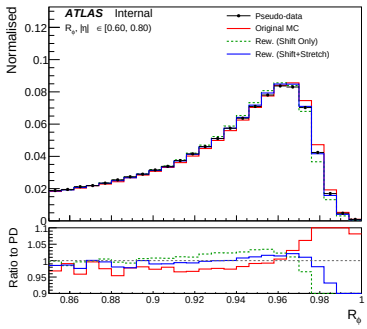


(c) $\delta = 5\%$

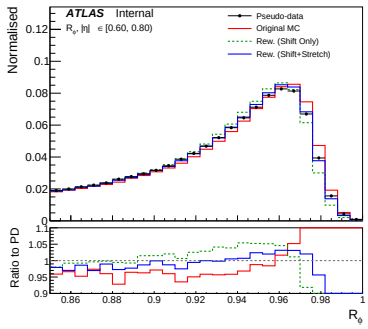
Figure 57: R_ϕ, $|\eta| \in [0.40, 0.60)$.



(a) $\delta = 1\%$

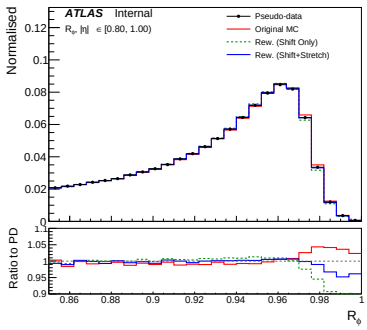


(b) $\delta = 3\%$

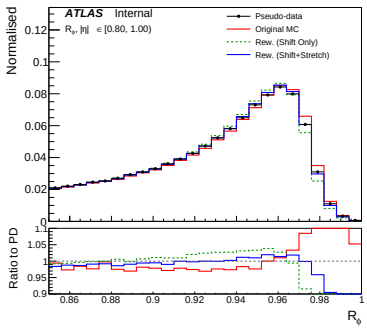


(c) $\delta = 5\%$

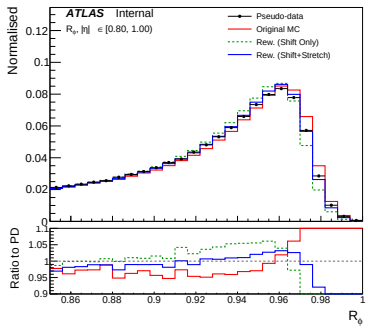
Figure 58: R_ϕ, $|\eta| \in [0.60, 0.80)$.



(a) $\delta = 1\%$

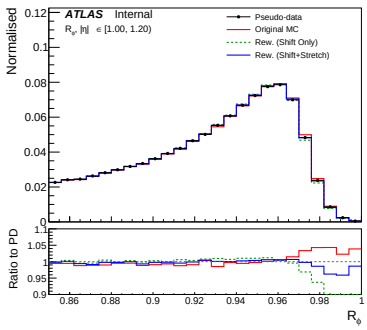


(b) $\delta = 3\%$

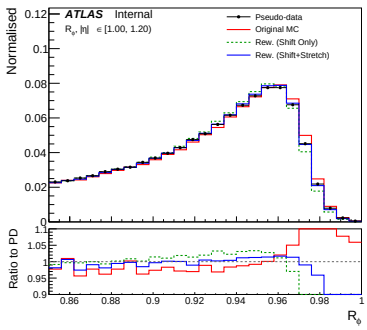


(c) $\delta = 5\%$

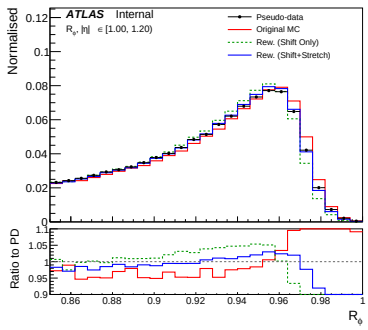
Figure 59: R_ϕ, $|\eta| \in [0.80, 1.00)$.



(a) $\delta = 1\%$

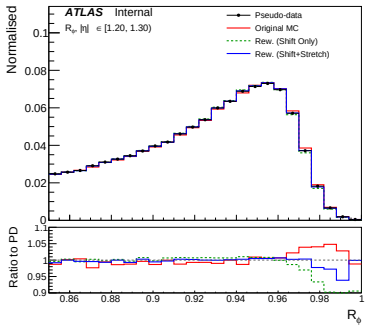


(b) $\delta = 3\%$

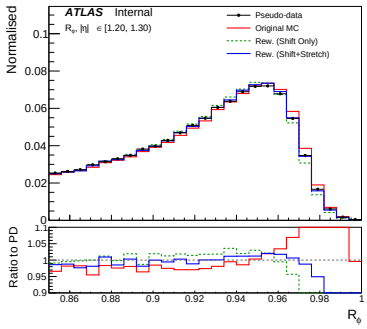


(c) $\delta = 5\%$

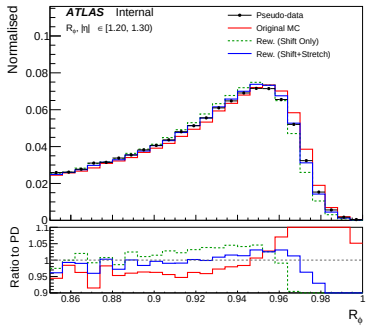
Figure 60: R_ϕ, $|\eta| \in [1.00, 1.20)$.



(a) $\delta = 1\%$

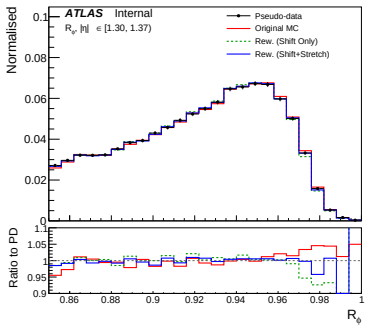


(b) $\delta = 3\%$

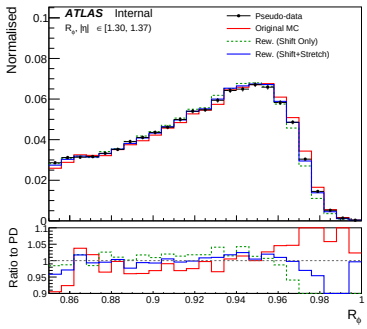


(c) $\delta = 5\%$

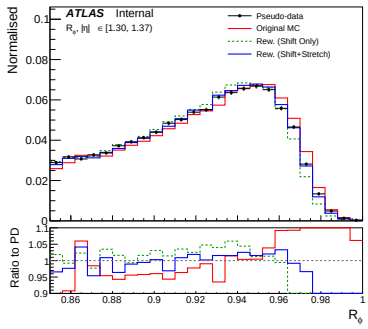
Figure 61: R_ϕ, $|\eta| \in [1.20, 1.30)$.



(a) $\delta = 1\%$

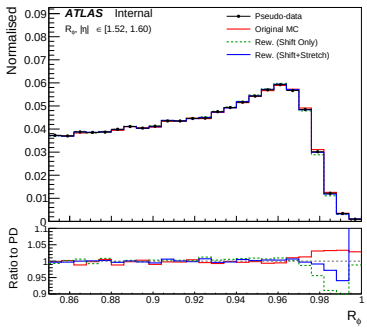


(b) $\delta = 3\%$

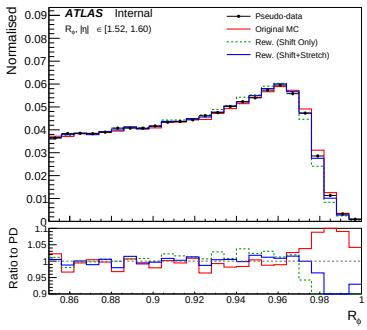


(c) $\delta = 5\%$

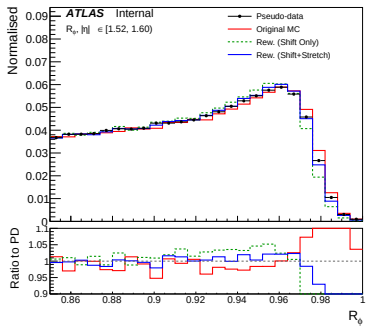
Figure 62: R_ϕ, $|\eta| \in [1.30, 1.37)$.



(a) $\delta = 1\%$

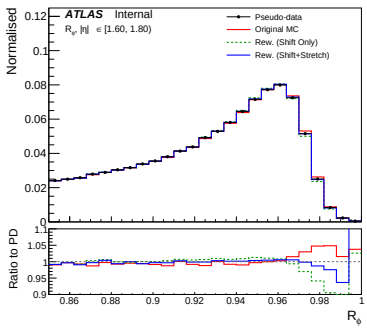


(b) $\delta = 3\%$

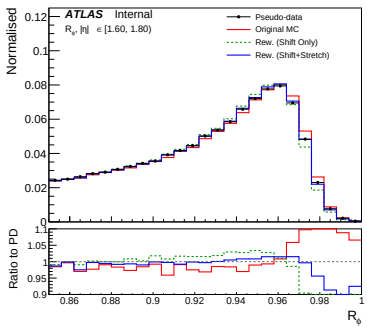


(c) $\delta = 5\%$

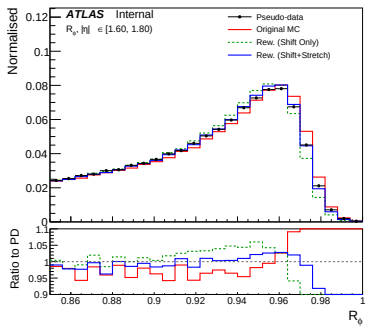
Figure 63: R_ϕ, $|\eta| \in [1.52, 1.60)$.



(a) $\delta = 1\%$

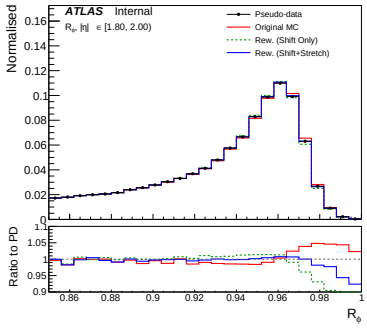


(b) $\delta = 3\%$

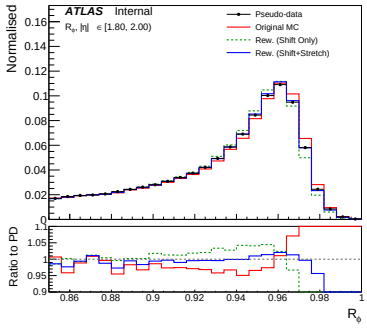


(c) $\delta = 5\%$

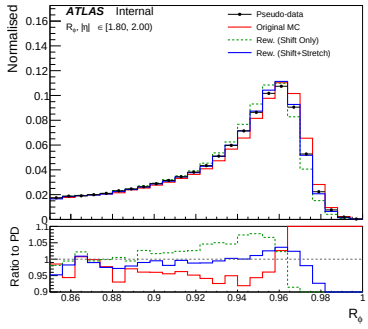
Figure 64: R_ϕ, $|\eta| \in [1.60, 1.80)$.



(a) $\delta = 1\%$

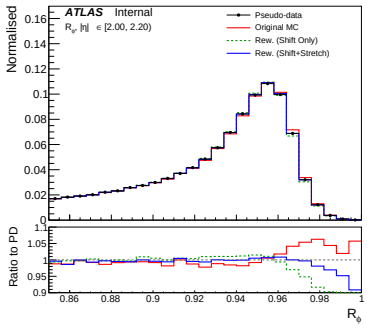


(b) $\delta = 3\%$

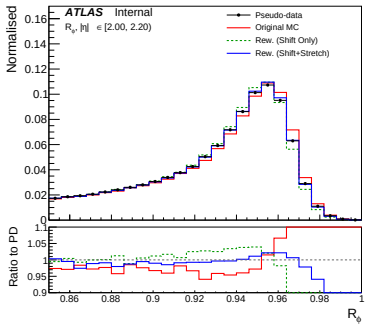


(c) $\delta = 5\%$

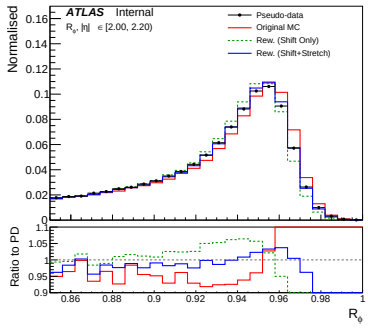
Figure 65: R_ϕ, $|\eta| \in [1.80, 2.00)$.



(a) $\delta = 1\%$

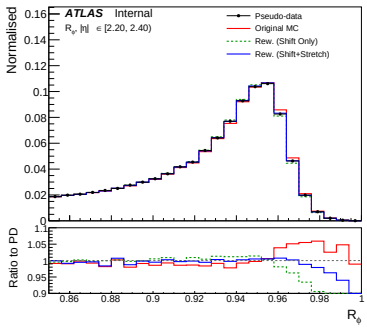


(b) $\delta = 3\%$

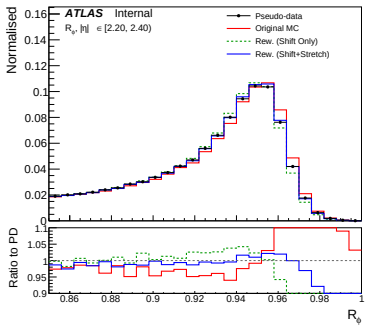


(c) $\delta = 5\%$

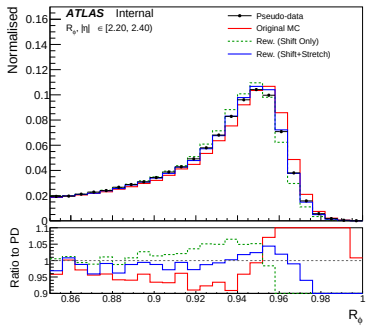
Figure 66: R_ϕ, $|\eta| \in [2.00, 2.20)$.



(a) $\delta = 1\%$



(b) $\delta = 3\%$



(c) $\delta = 5\%$

Figure 67: R_ϕ, $|\eta| \in [2.20, 2.40)$.